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Heating-Cooling Asymmetry in Quantum Systems

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Primeira frase do agradecimento ....

Segunda frase ....

Outras frases ....

Última frase ....

"I been a sucker ever since I was a baby
As a daughter, as a mother, as a lover, as a dear
But I'll fight them as a woman, not a lady
I'll fight them as [a physicist]!"
- Peggy Seeger (adapted)

ABSTRACT

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It is known that classical thermodynamics fails to explain the thermal relaxation of far-from-equilibrium systems. Recent studies have reported an asymmetry in this regime, where heating is faster than cooling for thermodynamically equidistant initial states. Understanding how quantum systems equilibrate is a fundamental question with implications for quantum technologies, such as state preparation for quantum computing. However, how quantum resources affect this asymmetry is an important open question. In this work, we investigate the heating and cooling asymmetry in discrete and continuous-variable quantum systems. For the discrete case, we study two interacting qubits coupled to a global bosonic bath, focusing on the effects of initial correlations and local non-Markovian dynamics. For the continuous case, we consider Gaussian states and compare the thermal relaxation of displaced and squeezed states. Our results for the discrete case indicate that non-Markovianity may accelerate thermal relaxation, while the initial coherences do not change the equilibration time. In Gaussian systems, applying the displacement operator in the initial states does not affect the asymmetry, but using initially squeezed states can reduce it.

Keywords: Asymmetry, Relaxation, Two-Interacting Qubits, Gaussian Systems

RESUMO

RAUPP, C. **Heating-Cooling Asymmetry in Quantum Systems**. 2026. 82 p. Dissertação (Mestrado em Ciências) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2026.

A termodinâmica clássica falha ao tentar explicar a relaxação térmica de sistemas fora do equilíbrio. Estudos recentes mostram uma assimetria nesse regime, onde esquentar é mais rápido do que resfriar estados iniciais termodinâmicamente equidistantes...

Palavras-chave: LaTeX. Classe USPSC. Tese. Dissertação. Trabalho de conclusão de curso (TCC). Relatório.

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1 INTRODUCTION

> If physical theories were people, thermodynamics would be the village witch. [...] The other theories find her somewhat odd, somehow different in nature from the rest, yet everyone comes to her for advice, and no-one dares to contradict her.
> - Goold, Huber, Riera, del Rio, Skrzypczyk (2016)

The reason everyone comes to this “village witch" for advise it is her pragmatic approach to nature. Instead of trying to describe microscopic details of a system, she focus on what can be done with its macroscopic resources. This is a consequence of thermodynamics being developed in the context of thermal machines and the goal to achieve efficient exploitation of the natural world. Moreover, this theory development brought fundamental laws regarding energy flow and transformation that have been used in different areas of physics, chemistry, biology, and other sciences. Despite thermodynamics describing macroscopic systems near equilibrium, there is a great effort to extend its concepts to nonequilibrium and quantum regime (1–4). This connection of quantum mechanics and thermodynamics can help to understand how quantum systems exchange energy with its surroundings. Because quantum properties can affect transient dynamics, relaxation processes of quantum systems become a non-trivial problem.

Relaxation processes are studied in different areas of physics. Some usual examples are resistor-capacitor circuit, damped harmonic oscillators, dielectric relaxation, and thermal relaxation. Despite their distinct contexts, this processes are connected in a general way with the question of how systems evolve in time after a perturbation. In this work, we will focus on thermal relaxation processes where we have system interacting with an environment. According to thermodynamics, thermal relaxation is the exchange energy varying the system’s temperature until it reaches a steady state. If the system is in a thermodynamic state near equilibrium, this evolution can be described using linear response theory, such as Onsager-Casemir and Kubo-Yokota-Nakajima laws (5). In this regime, the mean behaviour of statistical fluctuation is the same as the macroscopic system, being valid the assumption that the dynamics is memoryless and symmetric. Considering a thermal relaxation process, symmetry means that heating up or cooling down the system will take the same amount of time and follow the same thermodynamic trajectory.

However, thermal relaxation symmetry can break for systems initially far from equilibrium and quantum systems, where linear response theory fails. One example of thermal relaxation asymmetry is the Mpemba effect. First observations of the Mpemba effect was that a hotter system (such as water) would cool down faster than a colder one

(6, 7). There is also the inverse Mpemba effect that occurs in heating processes instead of cooling (7). In general terms, Mpemba-like effects consist of a system further from equilibrium relaxating faster than another system closer to the equilibrium. In this respect, Mpemba effect have been explored in the quantum regime (7, 8), such as the ergotropic Mpemba effect where it was found that squeezed states relaxates faster than displaced states even when the ergotropy of squeezing is bigger (which means it is further from equilibrium) (9). Another relaxation asymmetry that emerges in the far-from-equilibrium regime is the heating-cooling asymmetry. Recent experiments using Brownian motion showed that heating is faster than cooling (5, 10, 11). This was also observed to happen in quantum systems, such as a qubit, a quantum harmonic oscillator, and in quantum Brownian motion (12, 13). However, this result is not universal, being possible to observe cooling relaxating faster than heating depending on the system's characteristics, for example, energy gap or degeneracy in systems with more than two-levels (7,14,15). In addition, it is interesting that this heating-cooling asymmetry can also appears when comparing the usual Mpemba effect (cooling) with the inverse Mpemba effect (heating) (7).

A better understanding of relaxation asymmetries can be useful for state preparation, optimal transport, heat engines, and other applications (7). For this purpose, beyond phenomenological characterization, it is fundamental to study the physical mechanisms generating the asymmetries. To do this, previous studies developed an approach named thermal kinematics using the combination stochastic thermodynamics and information geometry (10, 16, 17). In thermal kinematics, thermodynamic states can be interpreted as points in a manifold, in which and we can define a distance between different states. As the system evolves in time, it will have a velocity and follows a thermodynamic trajectory in this manifold. Using these approach, Ref. (10) proved that heating and cooling systems far-from-equilibrium follow distinct thermodynamic trajectories. Later, thermal kinematics was extended to the quantum regime, where this same result was obtained (12). From a thermodynamic perspective, previous works showed that heating-cooling asymmetry can be explained by entropy production, in which heating is faster because it has a higher entropy production than cooling can dissipate heat (18). In addition to these works, understanding the consequences of quantumness in the heating-cooling asymmetry is still an open question. It is known that quantum correlations have consequences in thermodynamic processes, such as the case of anomalous heat exchange (19) and non-classical work extraction (20, 21). Additionally, in the context of thermal relaxation, coherence in a qubit was shown to delay equilibration (8). Therefore, the main question of this work is what quantumness do with the heating-cooling asymmetry.

However, there is a variety of systems and models we can use to explore this question. By observing different physical theories and phenomena, we see that discrete and continuous-variable models bring complementary explanations of nature. In addition, finding the analogous phenomena in both types of systems also enriches potential technological applications. Motivated by these considerations, this work explores the thermal relaxation asymmetry in both discrete and continuous-variable quantum systems. For the discrete case, we will focus on the thermal relaxation of two interacting qubits coupled to a global bath. Despite the variety of quantum properties we could investigate, we will focus only on quantum coherence and entanglement for this model. For the continuous-variable case, we focus on Gaussian states due to their relevance in quantum theory and experimental accessibility. The quantum resources explored in this case are caused by the displacement and squeezing of thermal states.

To study this question, we will use open quantum systems theory, in specific the Davies maps, which will be presented in Chap. 2. Then, in Chap. 3, we will ask advice to thermodynamics and extend some of its concepts and quantities, such as free energy and entropy, to the quantum regime. Also in this chapter, we introduce the thermal kinematics approach for both classical and quantum regime. Chap. 4 and Chap. 5 contains our analysis for the discrete and continuous-variable cases, respectively. In Chap. 6, we summarize our results and discuss future directions for this research.

2 QUANTUM SYSTEMS DYNAMICS

This chapter introduces how quantum systems evolve in time. In Sec. 2.1, we begin discussing the unitary evolution of closed quantum systems and argument how unitary dynamics fails to describe open quantum systems. Then, in Sec. 2.2, we will present the Markovian master equation approach and discuss how it can describe thermalisation processes in Sec. 2.2.1. The content of this chapter are based on Refs. (22–24).

2.1 Dynamics of Close Quantum Systems

Quantum mechanics was initially formulated for closed systems, whose physical states are represented as a normalised vector $|\psi\rangle$ in the Hilbert space $\mathcal{H}$. This representation contains all the information about the physical state, called a pure state. However, another mathematically equivalent formulation is more convenient for describing quantum systems whose states are not fully known. In these cases, it is only possible to assign a probability p_i of the system existing in the state $|\psi_i\rangle$, which is encapsulated in the density operator,

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i| \,. \tag{2.1}$$

The pure state is a special case in the density operator notation, where it is written as $\rho = |\psi\rangle \langle\psi|$. The density operator (also called density matrix) must satisfy the following conditions: (1) the sum of all p_i is equal to one, which is equivalent to $\mathrm{Tr}[\rho] = 1$, and (2) it is positive semi-definite, which means all ρ eigenvalues are $\lambda_k \geq 0$.

The time evolution of closed quantum systems is given by the Schrödinger equation,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle \,, \tag{2.2}$$

where $H(t)$ is the system's Hamiltonian and Planck's constant will be $\hbar = 1$ from now on. This equation can be translated into the density matrices notation as the Liouville-von Neumann equation,

$$\frac{d}{dt}\rho(t) = -i[H(t), \rho(t)]. \tag{2.3}$$

Its solution describes the unitary evolution of an initial state $\rho(0)$ as

$$\rho(t) = U(t)\rho(0)U^\dagger(t), \tag{2.4}$$

where $U(t)$ is a unitary operator with properties $U^{-1} = U^\dagger$ and $UU^\dagger = \mathbb{I}$, in which $\mathbb{I}$ is the identity matrix. As a consequence, the unitary dynamics preserve the information of the system, resulting in a time-reversible process.

Even though all this formalism describes a variety of quantum phenomena, it is an idealisation because every physical system interacts with an environment, which can change the system's dynamics. If we consider a quantum systems (S) interacting with an environment (E), we can assume that together they form a closed quantum system that evolves in time by the unitary dynamics described before. The total system ($S + E$) is defined by the Hamiltonian

$$H_{SE} = H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E + H_I, \tag{2.5}$$

where H_S is the system's Hamiltonian, H_E is the environment's Hamiltonian, and H_I describes the system-environment interaction. The environment is usually considered to have infinite degrees of freedom, over which we cannot access information or have control. As a result, the system S is the only part we can manipulate, being the focus of our studies. However, because the system S exchanges information with the environment, it does not have a unitary dynamics in general. This is what we call an open quantum system, illustrated in Fig. 1, whose dynamics will be introduced in the next section.

2.2 Dynamics of Open Quantum Systems

In this section, we discuss the fundamental steps to obtain the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation, also called the Lindblad master equation for short (22). We assume the time evolution of the total system ($S + E$) is unitary, meaning that the Liouville-von Neumann equation (Eq. (2.3)) can be used. However, we do not have access to all the information in the environment, and including its dynamics in the calculation can be overwhelming. Fortunately, it is possible to describe the system's dynamics while taking into account environmental effects, without needing to deal with the environment's time evolution.

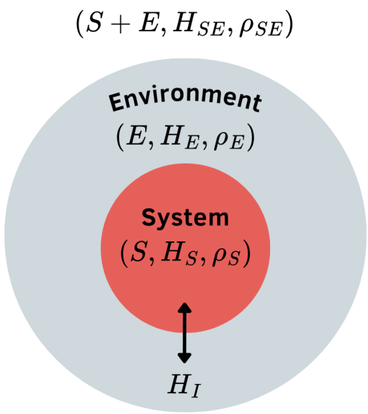

Figure 1 – A system S described by the Hamiltonian H_S is at a state ρ_S and it is coupled to an environment E described by the Hamiltonian H_E that is at a state ρ_E. They can interact according to Hamiltonian H_I and together they form a close system $S + E$ described by the Hamiltonian H_{SE} at a state ρ_{SE}.

Up to this point, we have used the Schrödinger picture, in which the state evolves with time while the observables remain unchanged. In this section, we will change to the interaction picture, where the Lindblad master equation is easer to deduce. The interaction picture have both states and operators changing with time (22, 25). To convert from the Schrödinger picture to the interaction picture, we split the total Hamiltonian into a time-independent and a time-dependent parts as $H = H_0 + V(t)$. Then, we define the interaction picture unitary operator $U_I(t) = U_0^\dagger(t)U(t)$ where $U(t)$ is the time-evolution operator of the Schrödinger picture discussed before (Eq. (2.4)) and $U_0(t) = \exp[-iH_0 t]$ (22,25). This allow us to write the state time-evolution in the interaction picture as

$$|\psi(t)\rangle_I = U_0^\dagger(t)\,|\psi(t)\rangle = U_I(t)\,|\psi(0)\rangle\,, \tag{2.6}$$

which is equivalent as writing the time-evolution of the density matrix as

$$\rho_I(t) = U_0^\dagger(t)\rho(t)U_0(t) = U_I(t)\rho(0)U_I^\dagger(t). \tag{2.7}$$

For observables, the time-evolution become

$$O_I(t) = U_0^\dagger(t)OU_0(t). \tag{2.8}$$

As a result, the Schrödinger equation becomes

$$i\frac{d}{dt}\,|\psi(t)\rangle_I = V_I(t)\,|\psi(t)\rangle_I\,, \tag{2.9}$$

where $V_I(t) = U_0^\dagger(t)V(t)U_0(t)$ is the time-dependent part of the Hamiltonian in the interaction picture (22, 25). Additionally, this result can be extended to the density matrices, resulting in the interaction picture Liouville-von Neumann equation,

$$\frac{d}{dt}\rho(t) = -i[V_I(t), \rho(t)]. \tag{2.10}$$

With this picture changes, we can go back to the open quantum system discussion. Our time-independent Hamiltonian is $H_0 = H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E$ and the time-dependent is the interaction between the system and the environment, which in the interaction picture is $V_I(t) = H_I(t)$. So, because $S + E$ is considered a closed system, their Liouville-von Neumann equation becomes

$$\frac{d}{dt}\rho_{SE}(t) = -i[H_I(t), \rho_{SE}(t)]. \tag{2.11}$$

We can integrate this equation in time and substitute into itself, obtaining

$$\frac{d}{dt}\rho_{SE}(t) = -i[H_I(t), \rho_{SE}(0)] - \int_0^t ds[H_I(t), [H_I(s), \rho_{SE}(s)]]. \tag{2.12}$$

However, this equation still takes account of the environment's time evolution and degrees of freedom, which can be overwhelming to solve as previously discussed. To solve this issue, we will make some assumptions and approximations to describe only the system's dynamics including environmental effects. First, we will assume that the initial system-environment state is separable, meaning they are not correlated: $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$. Next, we can do the Born approximation that consists of considering a weak interaction between the system and the environment, which keeps the global state ($\rho_{SE}(t)$) separable throughout the evolution. Given that the environment have much more degrees of freedom than the system, we can assume that the weak effects of the system will not change the environment's state. As a result, the environment's state is constant in time (ρ_E). Additionally, because we are only interested in the system, we take the partial trace over the environment, resulting in

$$\frac{d}{dt}\rho_S(t) = -i\mathrm{Tr}_E[H_I(t), \rho_S(0) \otimes \rho_E] - \int_0^t ds \mathrm{Tr}_E[H_I(t), [H_I(s), \rho_S(s) \otimes \rho_E]]. \tag{2.13}$$

We will also consider that the correlation time of the environment with itself is much shorter than the system relaxation time. In other words, this means that any perturbation of the system on the environment will be rapidly dissipated, resulting in a memoryless dynamics. This is called the Markovian approximation and guarantees that the system can only lose information to the environment. The absence of memory means that future states of the system do not depend on past states, so we use $\rho_S(s) = \rho_S(t)$. The last step to obtain a Markovian master equation is to make the dynamics independent of the system's initial state. This can be achieved by substituting $s \to t - s$ and extending $t \to \infty$. Thus, we find the Redfield equation,

$$\frac{d}{dt}\rho_S(t) = -i\mathrm{Tr}_E[H_I(t), \rho_S(0) \otimes \rho_E] - \mathrm{Tr}_E \int_0^\infty ds [H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_E]]. \tag{2.14}$$

Although Eq. (2.14) can describe the relaxation of open quantum systems, it does not guarantee the positivity of probabilities, which can lead to unphysical results. A solution is to use the secular approximation. In this section, we will study a simpler scenario, but the general case can be found in (22). Consider that a perturbation of the bath set off a single effect on the system, that is, no multiple transitions can occur simultaneously in the system. So, the interaction Hamiltonian in the Schrödinger picture can be decomposed into two Hermitian operators as

$$H_I = A \otimes B, \tag{2.15}$$

The next step is to write A in terms of the eigenstates of the system's Hamiltonian, $H_S |n\rangle = \lambda_n |n\rangle$. To make the secular approximation, we will separate this operator as $A = \sum_\omega A(\omega)$, where

$$A(\omega) = \sum_{\substack{n,m \\ \omega=\lambda_m-\lambda_n}} |n\rangle \langle n| A |m\rangle \langle m| \tag{2.16}$$

is summing over all combinations of states that have transition frequency ω. In the interaction picture, this Hamiltonian becomes

$$H_I(t) = \sum_{\omega} e^{-i\omega t} A(\omega) \otimes B(t) = \sum_{\omega'} e^{i\omega' t} A^\dagger(\omega') \otimes B^\dagger(t), \tag{2.17}$$

where the last equality follows the property $A(\omega) = A^\dagger(-\omega)$ and $\omega' = -\omega$. With this decompositions, we can write the interaction Hamiltonians from Eq. 2.14 as

$$H_I(t-s) = \sum_{\omega} e^{-i\omega(t-s)} A(\omega) \otimes B(t-s), \tag{2.18}$$

$$H_I(t) = \sum_{\omega'} e^{i\omega' t} A^\dagger(\omega') \otimes B^\dagger(t). \tag{2.19}$$

When we use these definitions, we will have that the first term in Redfield equation (Eq. (2.14)) is

$$-i\mathrm{Tr}_E[H_I(t), \rho_S(0) \otimes \rho_E] = -i \sum_{\omega'} e^{-i\omega' t} [A^\dagger(\omega'), \rho_S(0)] \langle B^\dagger(t) \rangle_E \,, \tag{2.20}$$

where $\langle B^\dagger(t) \rangle_E = \mathrm{Tr}[B^\dagger(t)\rho_E]$ is the one-point correlation function. Assuming the environment is at a steady state, we have $\langle B^\dagger(t) \rangle_E = 0$ and the first term in Eq. (2.14) disappears (26, 27). Thus, we obtain

$$\begin{aligned} \frac{d}{dt}\rho_S(t) &= -\mathrm{Tr}_E \int_0^\infty ds [H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_E]] \\ &= \sum_{\omega,\omega'} e^{i(\omega'-\omega)t} \Gamma(\omega)(A(\omega)\rho_S(t)A^\dagger(\omega') - A^\dagger(\omega')A(\omega)\rho_S(t)) + \text{h.c}, \end{aligned} \tag{2.21}$$

where

$$\Gamma(\omega) = \int_0^\infty ds e^{i\omega s} \langle B^\dagger(t) B(t-s) \rangle_E \,, \tag{2.22}$$

with $\langle B^\dagger(s) B(0) \rangle_E = \mathrm{Tr}[B^\dagger(s)B(0)\rho_E]$ being the two-point correlation functions of the bath. The secular approximation imposes that the terms with $\omega \neq \omega'$ oscillate so fast that they do not affect the relaxation process. As a result, only the resonance case ($\omega = \omega'$) contribute to the equation. Thus, we have

$$\frac{d}{dt}\rho_S(t) = \sum_{\omega} \Gamma(\omega)(A(\omega)\rho_S(t)A^\dagger(\omega) - A^\dagger(\omega)A(\omega)\rho_S(t)) + \text{h.c.} \tag{2.23}$$

One last step to obtain the usual form of the Lindblad master equation is to separate $\Gamma(\omega)$ into its real and imaginary parts as

$$\Gamma(\omega) = \frac{1}{2}\gamma(\omega) + iS(\omega), \tag{2.24}$$

where the transition rate is

$$\gamma(\omega) = \int_{-\infty}^\infty ds e^{i\omega s} \langle B^\dagger(s) B(0) \rangle_E \,, \tag{2.25}$$

and $S(\omega)$ produce slight changes on the system's energy levels, which are included in the Lamb shift Hamiltonian

$$H_{LS} = \sum_{\omega} S(\omega) A^{\dagger}(\omega) A(\omega). \tag{2.26}$$

As a result, the Lindblad master equation in the interaction picture has the form

$$\frac{d}{dt}\rho_S(t) = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)), \tag{2.27}$$

with the dissipator being

$$\mathcal{D}(\rho_S) = \sum_{\omega} \gamma(\omega) \left(A(\omega)\rho_S A^{\dagger}(\omega) - \frac{1}{2}\{A^{\dagger}(\omega)A(\omega), \rho_S\} \right), \tag{2.28}$$

where $A(\omega)$ are usually called jump operators. In the Schrödinger picture, the Lindblad master equation becomes

$$\frac{d}{dt}\rho_S(t) = -i[H_S + H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)), \tag{2.29}$$

where the jump operators remain the same. It is very common to neglect the Lamb shift Hamiltonian, because it only changes the energy gap of the system without changing the Hamiltonian structure, which preserves the system's physics. That said, from now on, we will use only the H_S in our Lindblad master equation.

2.2.1 Davies Maps

An important class of Lindblad master equations is the Davies maps, which generate the thermalisation of a quantum system weakly coupled to a thermal bath at inverse temperature β (28). The steady state of the Davies maps is always an equilibrium thermal state (also called Gibbs state), that is defined as

$$\rho_{ss} = e^{-\beta H_S}/\mathcal{Z}_S, \tag{2.30}$$

where $\mathcal{Z}_S = \mathrm{Tr}[\exp(-\beta H_S)]$ is the partition function. An important characteristic of this dynamics is that it represent processes that obey the detailed balance condition, meaning that the transition rates $\gamma(\omega)$ (Eq. (2.25)) must satisfy

$$\gamma(-\omega) = \gamma(\omega)e^{-\beta\omega}. \tag{2.31}$$

Given these conditions, the jump operators of the Davies map have a specific format. Let's rewrite the dissipator (Eq. (2.28)) as

$$\mathcal{D}(\rho_S) = \sum_{\omega} L_{\omega}\rho_S L_{\omega}^{\dagger} - \frac{1}{2}\{L_{\omega}^{\dagger}L_{\omega}, \rho_S\}, \tag{2.32}$$

defining the jump operators as $L_\omega = \sqrt{\gamma(\omega)}A(\omega)$. To compute the transition rate ($\gamma(\omega)$), according to Eq. (2.25), we need to use that our environment is a thermal bath. The thermal bath can be described by a set of infinite independent harmonic oscillators, $H_E = \sum_k \omega_k b_k^\dagger b_k$, where ω_k is the frequency of the k-mode, and b_k and $b_k^\dagger$ are the annihilation and creation operators of the k-mode. That said, the hermitian operator from Eq. (2.15) is $B = \sum_k (g_k b_k^\dagger + g_k^* b_k)$, that in the interaction picture becomes

$$B(s) = \sum_k (g_k b_k^\dagger e^{i\omega_k s} + g_k^* b_k e^{-i\omega_k s}), \tag{2.33}$$

with g_k being the system-environment coupling parameter. The density matrix of the thermal bath is

$$\rho_E = \prod_i (1 - e^{-\beta\omega_i}) e^{-\beta\omega_i b_i^\dagger b_i}, \tag{2.34}$$

with which we can compute the correlation functions (defined in Eq. (2.25)), obtaining

$$\langle B^\dagger(s)B(0)\rangle_E = \sum_k |g_k|^2 (e^{-i\omega_k s}(\bar{n}(\omega_k) + 1) + e^{i\omega_k s}\bar{n}(\omega_k)), \tag{2.35}$$

where $\bar{n}(\omega_k) = (e^{\beta\omega_k} - 1)^{-1}$ is the Bose-Einstein distribution. Then, we find the transition rates by solving the integral in Eq. (2.25), resulting in

$$\gamma(-\omega) = 2\pi \sum_k |g_k|^2 \bar{n}(\omega_k)\delta(\omega - \omega_k) \tag{2.36}$$

$$\gamma(+\omega) = 2\pi \sum_k |g_k|^2 (\bar{n}(\omega_k) + 1)\delta(\omega - \omega_k). \tag{2.37}$$

Assuming that the infinite harmonic oscillators of the bath have close frequencies, we can convert from discrete to continuum space by transforming the sum into an integral in ω, g_k into a function $g(\omega)$ and including the density function of the bath's states $d(\omega)$. The solution of this new integral gives transition rates that respect the detailed balance (Eq. (2.31)),

$$\gamma(-\omega) = \Gamma(\omega)\bar{n}(\omega), \tag{2.38}$$

$$\gamma(+\omega) = \Gamma(\omega)(\bar{n}(\omega) + 1), \tag{2.39}$$

where $\Gamma(\omega) = 2\pi d(\omega)|g(\omega)|^2$. Considering a homogeneous bath and the same coupling for all k-modes, Γ becomes a constant. Also, because we have two frequencies, we solve the sum in ω from the Eq. (2.32), obtaining the following Lindblad equation for the Davies maps,

$$\frac{d}{dt}\rho_S(t) = -i[H_S, \rho_S(t)] + \mathcal{D}^-(\rho_S) + \mathcal{D}^+(\rho_S), \tag{2.40}$$

where each dissipator's jump operator is

$$L_\omega^+ = \sqrt{\Gamma(\bar{n}(\omega) + 1)}\, \langle n| A |m\rangle\, |n\rangle \langle m|\,, \tag{2.41}$$

$$L_\omega^- = \sqrt{\Gamma\bar{n}(\omega)}\, \langle m| A^\dagger |n\rangle\, |m\rangle \langle n|\,, \tag{2.42}$$

and we used the definition of $A(\omega) = A^\dagger(-\omega)$ in Eq. (2.16) and that $m < n$. The physical interpretation of these operators is that L^+_ω describes the absorption process (system gain energy) and the L^-_ω describes the spontaneous emission process (system loses energy).

After discussing some of the open quantum systems theory, it is always helpful to solve an example. Below, we will solve the thermalisation of one qubit and a single-mode harmonic oscillator, whose solution will be used later in this work's results.

2.2.1.1 Example: one qubit coupled to a thermal bath

One qubit can be described by the Hamiltonian $H_S = -\frac{\omega_0}{2}\sigma^z = -\frac{\omega_0}{2}(|0\rangle\langle 0| - |1\rangle\langle 1|)$. Consider this qubit is coupled to a thermal bath of inverse temperature β and evolve in time according to the Davies maps. Suppose the qubit can absorb energy from the bath as $\sigma^+|0\rangle = |1\rangle$ (where $\sigma^+ = |1\rangle\langle 0|$), or it can dissipate energy to the bath as $\sigma^-|1\rangle = |0\rangle$ (where $\sigma^- = |0\rangle\langle 1|$). Using this description, the Hermitian operator of the interaction effect on the system's subspace is defined as $A = \sigma^+ + \sigma^-$ and its decomposition becomes

$$A(+\omega_0) = |1\rangle\langle 1| A |0\rangle\langle 0| = \sigma^+, \tag{2.43}$$

$$A(-\omega_0) = |0\rangle\langle 0| A |1\rangle\langle 1| = \sigma^-. \tag{2.44}$$

As a result, the jump operators of one qubit thermalization are

$$L^+_\omega = \sqrt{\Gamma(\bar{n}(\omega)+1)}\sigma^+, \tag{2.45}$$

$$L^-_\omega = \sqrt{\Gamma\bar{n}(\omega)}\sigma^-. \tag{2.46}$$

2.2.1.2 Example: single-mode harmonic oscillator coupled to a thermal bath

Consider the system is a single-mode harmonic oscillator $H_S = \omega_0 a^\dagger a$ with a and $a^\dagger$ being its annihilation and creation operators, respectively. As we discussed before, this system is coupled to a thermal bath of temperature β, and the relaxation process is described by Eq. (2.40).

To find out the jump operators of this case, we define $A = a + a^\dagger$ as the Hermitian operator of the interaction acting on the system subspace. To decompose A into $A(\omega)$, we must find out the eigenstates of H_S and the frequencies at which this interaction operator makes transitions in our systems. The eigenstates of the harmonic oscillator are Fock states, $H_S|n\rangle = \lambda_n|n\rangle$ with $\lambda_n = n\omega_0$. For these states, the creation operator raises one energy level ($a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$) and the annihilation operator lowers one energy level ($a|n\rangle = \sqrt{n}\,|n-1\rangle$) Using that the energy difference is given by $\lambda_m - \lambda_n = (m-n)\omega_0$, the possible transition frequencies are $+\omega_0$ or $-\omega_0$. Then, the terms of the decomposition of A according to Eq. (2.16) becomes

$$A(+\omega_0) = \sum_m |m-1\rangle \langle m-1| A |m\rangle \langle m| = a, \tag{2.47}$$

$$A(-\omega_0) = \sum_m |m+1\rangle \langle m+1| A |m\rangle \langle m| = a^\dagger. \tag{2.48}$$

As a result, the jump operators of this thermal relaxation are

$$L_\omega^+ = \sqrt{\Gamma(\bar{n}(\omega)+1)}a, \tag{2.49}$$

$$L_\omega^- = \sqrt{\Gamma\bar{n}(\omega)}a^\dagger. \tag{2.50}$$

This example is fundamentally related to Gaussian states, which will be introduced in Chap. 5. In Sec. 5.2, we will analytically solve this master equation.

3 QUANTUM THERMODYNAMICS

In the last chapter, we discussed how to describe the time evolution of quantum systems, with open quantum systems dynamics being the most interesting to us. The system evolves in time from a nonequilibrium state to a stationary state (which is a thermal state for Davies maps) through exchanges of energy and information with an environment. All of this was discussed through the lens of quantum mechanics, but it clearly has a strong connection with thermodynamics. In this chapter, we will explore how thermodynamics appears in the context of open quantum systems. The main question we will address in Sec. 3.1 is how classical thermodynamic variables and laws emerge from the quantum theory. This is particularly important because thermodynamics is known as the theory that tells what nature is allowed to do in energy terms, so extending it to the quantum regime can bring new and deeper understanding. Additionally, in Sec. 3.2, we will discuss how information theory and thermodynamics are connected using the geometric interpretation of relaxation processes, which originated the thermal kinematics.

3.1 Thermodynamic variables in the quantum regime

In a general sense, thermodynamics is a theory about how energy changes. According to the first law, energy can be transferred by heat and work. These concepts define heat as the energy exchanged for reconfiguration of the system's state and is uncontrollable, while work is the energy that can be exploited and controlled. In quantum mechanics, the average energy of any system ρ with Hamiltonian H is defined as $E(\rho) = \mathrm{Tr}[\rho H]$, but its decomposition into heat and work is not unique (29–31). For this reason, in this work, we will use an alternative approach to study the thermodynamics of quantum systems. The entropic approach not only avoid the ambiguities of heat and work definitions, but also allows a connection with information theory.

In Chap. 2, we have already named some processes as reversible or irreversible, which can be explained and quantified by the second law of thermodynamics. This law states that the entropy of an isolated system can only increase (irreversible energy exchange) or be conserved (reversible energy exchange). However, there are many different definitions for entropy. In classical thermodynamics, entropy accounts for all the possible microstates Ω given some macroscopic constraints: $S = k_B \ln \Omega$, being k_B the Boltzmann constant (32). In other words, this entropy measures our lack of information about the system's microstate. In addition to that, information theory introduces a convenient interpretation of entropy. Consider a random variable X with probability distribution p_i. To know how much information is contained in the random variable outcomes, we can use the Shannon entropy $H(X) = -\sum_i p_i \log_2 p_i$, bounded by zero when the outcome is absolutely certain (we have all information), and being maximum when all outcomes are equally likely (we have no information) (33). Despite thermodynamic and Shannon entropies measuring “ignorance", they are not naturally equivalent, but Shannon entropy can be considered a thermodynamic entropy when the probabilities follow the Boltzmann distribution (3, 34).

In the quantum regime, we use density operators instead of probability distributions, which leads us to another quantity that is the von Neumann entropy (33),

$$S(\rho) = -\mathrm{Tr}[\rho \ln \rho]. \tag{3.1}$$

The von Neumann entropy becomes equivalent to the Shannon entropy for diagonal states. Also, it is zero for pure states (because we know all of its information), and its maximum is $\log d$ for maximally mixed states, where d is the dimension of the Hilbert space (33). More than characterising a state, the von Neumann entropy brings the second law into the quantum regime. In Chap. 2, we discussed that unitary evolutions do not lose information, which allowed us to identify them as reversible. In other words, this means that unitary evolutions preserve the system's entropy. To show this, we can take the time derivative of the von Neumann entropy as

$$\begin{aligned}\frac{\partial S(\rho)}{\partial t} &= -\mathrm{Tr}\left[\frac{\partial \rho}{\partial t}\ln\rho + \rho\frac{\partial}{\partial t}\ln\rho\right]\\ &= -\mathrm{Tr}\left[\frac{\partial \rho}{\partial t}\ln\rho + \frac{\partial \rho}{\partial t}\right]\\ &= i\mathrm{Tr}\left[[H,\rho]\ln\rho\right]\\ &= i\left(\mathrm{Tr}[H\rho\ln\rho] - \mathrm{Tr}[\rho H\ln\rho]\right)\\ &= 0,\end{aligned} \tag{3.2}$$

where we used that $\frac{\partial}{\partial t}\ln\rho = \rho^{-1}\frac{\partial \rho}{\partial t}$ and in the last line we used that $[\rho, \ln\rho] = 0$. However, this is not valid for open quantum systems, because information can be exchanged and lost. From the perspective of the system under a Markovian dynamics, its entropy can only increase respecting the second law of thermodynamics. To quantify how much entropy is produced in irreversible processes, we can separate the entropy variation into a flux between the system and the environment (that is connected with Clausius theorem) and an entropy production (Π_ρ) (35). By assuming the environment is at temperature β^{-1}, the flux becomes $\beta\delta Q$, where $\delta Q = \mathrm{Tr}[\dot{\rho}H]$ is the heat flow between system and environment (35). As a consequence, the balance equation for the entropy becomes

$$\frac{\partial S}{\partial t} = \beta\delta Q + \Pi_\rho, \tag{3.3}$$

where S is the von Neumann entropy. Given a thermal state $\rho_{th} = \exp(-\beta H)/\mathcal{Z}$, we have that $\ln\rho_{th} = -\beta H - \ln\mathcal{Z}$. By considering the equilibrium state is a thermal state, we can write that

$$\begin{aligned}\beta\delta Q &= -\mathrm{Tr}[\dot{\rho}(\ln\rho_{\mathrm{eq}} + \ln\mathcal{Z})] \\ &= -\frac{\partial}{\partial t}(\mathrm{Tr}[\rho\ln\rho_{\mathrm{eq}} + \rho\ln\mathcal{Z}]) \\ &= -\frac{\partial}{\partial t}\mathrm{Tr}[\rho\ln\rho_{\mathrm{eq}}],\end{aligned} \tag{3.4}$$

where we used that both ρ_{eq} and $\ln\mathcal{Z}$ are constant in time. Then we can find that the entropy production becomes

$$\begin{aligned}\Pi_\rho &= \frac{\partial S}{\partial t} - \beta\delta Q \\ &= -\frac{\partial}{\partial t}(\mathrm{Tr}[\rho\ln\rho] - \mathrm{Tr}[\rho\ln\rho_{\mathrm{eq}}]) \\ &= -\frac{\partial}{\partial t}D[\rho||\rho_{\mathrm{eq}}]\end{aligned} \tag{3.5}$$

where

$$D[\rho||\sigma] = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)], \tag{3.6}$$

is the relative entropy, which measures the distinguishability between two states. An interesting interpretation of Eq. (3.5) is that entropy production measures the speed that the system converges to equilibrium (18). So higher entropy production results in faster relaxation. Additionally, it is possible to show that the entropy production can only be greater than zero for irreversible processes, as stated by the second law of thermodynamics. In the context of Lindblad master equations, the relative entropy decreases monotonically as the system evolve in time *, which explains the positivity of the entropy production (36).

Another important aspect about Eq. (3.5) is that the relaxation process depends on how far from equilibrium the system is, because the relative entropy measures how distinguishable the system's state is from the equilibrium state. However, the relative entropy is not a proper measure of distance, because it is not symmetric ($D[P||Q] \neq D[Q||P]$) and do not satisfy triangle inequality, but it can be used to estimate how much useful energy is available in a nonequilibrium state. To see this, we can rewrite the average energy by introducing the equilibrium state $\rho_{eq} = \rho_{th}$ as

$$\begin{aligned}E(\rho) = \mathrm{Tr}[\rho H] &= -\beta^{-1}\mathrm{Tr}[\rho\ln e^{-\beta H}] \\ &= -\beta^{-1}\mathrm{Tr}\left[\rho\ln\frac{e^{-\beta H}}{\mathcal{Z}} + \rho\ln\mathcal{Z}\right] \\ &= -\beta^{-1}\mathrm{Tr}\left[\rho\ln\rho_{\mathrm{eq}}\right] - \beta^{-1}\ln\mathcal{Z} + \beta^{-1}S(\rho) - \beta^{-1}S(\rho) \\ &= \beta^{-1}D[\rho||\rho_{\mathrm{eq}}] + \beta^{-1}S(\rho) + F_{\mathrm{eq}},\end{aligned} \tag{3.7}$$

* This only occurs for dynamics described by completely positive trace-preserving maps (CPTP), such as the Lindblad master equation. Otherwise, there is no guarantee for the positivity of entropy production (36), as will be discussed in Sec. 4.3.

where $S(\rho)$ is the von Neumann entropy and $F_{\rm eq} = -\beta^{-1}\ln\mathcal{Z}$ is the equilibrium free energy (8). The free energy definition $F = E(\rho) - \beta^{-1}S(\rho)$ is an extension of the Helmholtz free energy (3), and can be used to find

$$F_{\rm neq} = \beta^{-1}D[\rho||\rho_{\rm eq}] + F_{\rm eq}, \tag{3.8}$$

that is the nonequilibrium free energy. This expression shows that the relative entropy is directly related to the amount of energy a state has because it is far from equilibrium. It is interesting to observe that the nonequilibrium free energy is related to the entropy production as $\Pi_\rho = -\beta\frac{\partial}{\partial t}F_{\rm neq}$. Also, this energetic interpretation can also be used to find that the maximum amount of energy that could be extracted from the system,

$$W_{\rm max} = \Delta F = F_{\rm neq} - F_{\rm eq}. \tag{3.9}$$

However, this equality is only satisfied when the final state after work extraction is a thermal state, which is a consequence of thermal states having the minimum possible energy (37). Thermal states are just one type of what we call passive states (represented as π), which are defined as states from which no work can be extracted via cyclic unitary operations: $\mathrm{Tr}[H\pi] \leq \mathrm{Tr}[HU\pi U^\dagger]$ (21). Is important to emphasise that the energy of any passive state is $E(\pi) \geq E(\rho_{th})$. From the definition of passivity, we can also define the maximum work extracted via cyclic unitary operations as

$$\mathcal{E}(\rho) = E(\rho) - E(\pi), \tag{3.10}$$

where the energy of the passive state is $E(\pi) = \min_U \mathrm{Tr}[HU\rho U^\dagger]$ (9, 38). Because we are considering unitary operations, the von Neumann entropy is conserved ($S(\rho) = S(\pi)$) and we can use the extended Helmholtz free energy to rewrite $W_{\rm max} = E(\rho) - E(\rho_{\rm eq})$, where $\rho_{\rm eq}$ is a thermal state. Now, the comparison of ergotropy with the maximum work becomes $\mathcal{E}(\rho) \leq W_{\rm max}$, with equality for $\pi = \rho_{th}$ (21, 37).

Ergotropy can also be used to understand entropy production. To see this, we can rewrite Eq. (3.10) using the definition of energy in Eq. (3.7) as

$$\begin{aligned}\mathcal{E}(\rho) &= \beta^{-1}D[\rho||\rho_{eq}] + \beta^{-1}S(\rho) + F_{\rm eq} - \beta^{-1}D[\pi||\rho_{eq}] - \beta^{-1}S(\pi) - F_{\rm eq} \\ &= \beta^{-1}(D[\rho||\rho_{eq}] - D[\pi||\rho_{eq}]).\end{aligned} \tag{3.11}$$

Now, by making the time derivative of this expression for ergotropy, we can find that

$$\Pi_\rho = \Pi_\pi - \beta^{-1}\dot{\mathcal{E}}(\rho), \tag{3.12}$$

which means the entropy production can be separated into a passive contribution and an ergotropic contribution where $\dot{\mathcal{E}}(\rho)$ is called ergotropic loss rate. A better discussion of this separation is done in Sec. 5.2, where it will be useful for the heating-cooling asymmetry in Gaussian states.

3.2 Thermal Kinematics

In previous sections of this chapter, we discussed how thermodynamic and informational quantities behave during relaxation processes. Furthermore, we can study relaxation processes through another approach that is a combination of stochastic thermodynamics and informational geometry, named thermal kinematics (10). In this approach, each physical state is represented as a point in a Riemann space, that is a curve surface also called manifold (16, 39). As a result, the time evolution of states become trajectories in this manifold, enabling the definition of quantities with analogous interpretation from kinematics, such as length and velocity (16, 17). Initially, thermal kinematics was originated to discuss far-from-equilibrium relaxation processes of classical systems (10), and later, it was extended to the quantum regime (12).

For convenience, we being with the discussion of the classical case. The central piece for connecting thermodynamics with information geometry is a quantity called Fisher information, that measures how much information we gain about a system by varying its parameters (40,41). Given that a classical system can be described by the probability distribution $P(x,\theta)$ of being in state x with the parameter θ, the classical Fisher information is

$$\mathcal{I}_c(\theta) = \int_{-\infty}^{\infty} \frac{(d_\theta P(x,\theta))^2}{P(x,\theta)} dx, \tag{3.13}$$

where d_θ is the derivative with respect to θ. To give a geometrical interpretation for the Fisher information, we can use that each thermodynamic state associated with the $P(x,\theta)$ is a point in a statistical manifold. As a result, if we have distinguishable states, they are different points in this statistical space, enabling the definition of a statistical distance between them (17,39). As previously discussed in Sec. 3.1, one measure of distinguishability is the relative entropy, that in term of probability distributions have the form

$$D[F(x)||Q(x)] = \int dx F(x) \ln\left(\frac{F(x)}{Q(x)}\right). \tag{3.14}$$

Consider an initial state with probability distributions $P(x,\theta)$ that evolve in an infinitesimal time dt becoming $P(x,\theta+d\theta)$. Because they are distinct probability distributions, it means they are separated by a infinitesimal distance ds in the statistical manifold. To distinguish the probabilities distributions, we can use the relative entropy and make a second order Taylor expansion of $P(x,\theta+d\theta)$ to obtain

$$\begin{aligned}
D[P(x,\theta+d\theta)||P(x,\theta)] &= \int dxP(x,\theta+d\theta)\ln\left(\frac{P(x,\theta+d\theta)}{P(x,\theta)}\right) \\
&= \int dx[P(x,\theta)+\partial_\theta P(x,\theta)d\theta+\frac{1}{2}\partial_\theta^2 P(x,\theta)d\theta^2+\mathcal{O}(d\theta^3)]\times \\
&\times\left[\frac{\partial_\theta P(x,\theta)}{P(x,\theta)}d\theta+\frac{1}{2}\frac{\partial_\theta^2 P(x,\theta)}{P(x,\theta)}d\theta^2+\mathcal{O}(d\theta^3)\right] \\
&= \int dx\left[\partial_\theta P(x,\theta)d\theta+\frac{1}{2}\partial_\theta^2 P(x,\theta)d\theta^2+\frac{(\partial_\theta P(x,\theta)d\theta)^2}{P(x,\theta)}+\mathcal{O}(d\theta^3)\right] \\
&= \int dx\frac{(\partial_\theta P(x,\theta)d\theta)^2}{P(x,\theta)}+\mathcal{O}(d\theta^3) \\
&= \mathcal{I}_c(\theta)d\theta^2+\mathcal{O}(d\theta^3)
\end{aligned} \tag{3.15}$$

where we used $\ln(1+x)=x+\mathcal{O}(x^2)$ in the second equality and $\int dxP(x,\theta)=1$ in the fourth equality (10). From this, we identify the Chentsov's theorem as $ds^2=\mathcal{I}_c(\theta)d\theta^2$, which means the Fisher information is the unique Riemannian metric in this statistical manifold (16, 39, 42). In other words, the statistical line element depends on the Fisher information as $ds=\sqrt{\mathcal{I}(\theta)}d\theta$. In the context of thermodynamics, the parameter of interest in relaxation processes is time ($\theta=t$), which allow us to define the instantaneous speed (also called statistical velocity) of the relaxation process as

$$v_c(t)=\sqrt{\mathcal{I}(t)}, \tag{3.16}$$

and the statistical length between the two states as

$$\mathcal{L}_c(t)=\int_0^t v_c(\tau)d\tau=\int_0^t\sqrt{\mathcal{I}_c(\tau)}d\tau. \tag{3.17}$$

Additionally, we can define another quantity that is the degree of completion, which measures how much of the total trajectory until time t_f the system has completed at time t_s,

$$\varphi_c(t_s)=\frac{\mathcal{L}_c(t_s)}{\mathcal{L}_c(t_f)}, \tag{3.18}$$

being $\varphi_c\in[0,1]$. These three quantities describes the thermal kinematic for classical systems, being a important to understand relaxation processes in terms of thermodynamic trajectories.

However, quantum systems are described by density matrices instead of probability distributions, being necessary some adaptations in thermal kinematics derivation. First, we need to use the quantum Fisher information (QFI) instead of the definition in Eq. 3.13. The QFI can be defined as

$$\mathcal{I}_Q(\rho(t))=2\sum_{n,m}\frac{|\langle\psi_n|\,\partial_t\rho(t)\,|\psi_m\rangle|^2}{\lambda_n+\lambda_m}, \tag{3.19}$$

where λ_k and $|\psi_k\rangle$ are, respectively, the kth eigenvalue and eigenvector of the density matrix $\rho(t)$ (41,43). Nevertheless, this expression do not naturally appears from expanding the relative entropy, but it can be found using other distinguishability measure. The Bures distance measures the statistical distance between two density matrices,

$$[D_B(\rho,\sigma)]^2 = 2[1 - F(\rho,\sigma)], \tag{3.20}$$

where $F(\rho,\sigma) = \mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$ is the fidelity, that measures how similar two density matrices are. Similar to the relative entropy case, we make a Taylor expansion of $\rho(t+dt)$ up to the second order and after some algebra we find (41)

$$ds^2 = [D_B(\rho(t),\rho(t+dt))]^2 = \frac{1}{4}\mathcal{I}_Q(\rho(t))dt^2 + \mathcal{O}(dt^3). \tag{3.21}$$

From this equation, we can redefine the thermal kinematics variables as

$$v(t) = \frac{1}{2}\sqrt{\mathcal{I}_Q(\rho(t))} \tag{3.22}$$

$$\mathcal{L}(t) = \frac{1}{2}\int_0^t \sqrt{\mathcal{I}_Q(\rho(t))} \tag{3.23}$$

$$\varphi(t_s) = \frac{\mathcal{L}(t_s)}{\mathcal{L}(t_f)}. \tag{3.24}$$

4 HEATING-COOLING ASYMMETRY: DISCRETE CASE

Experiments showed that heating is faster than cooling in Langevin systems (10) and theoretical studies showed that this asymmetry also exist in quantum systems (12). However, understanding how quantum resources affect heating-cooling asymmetry is still an open question. In this chapter, we will investigate quantum coherence and entanglement effects on the asymmetry. For that, we consider two interacting qubits coupled to a global bosonic bath. However, before we discuss the results of this work, it is important to present the quantum correlations we will investigate, which is done in Sec. 4.1. Then, in Sec. 4.2, we examine characteristics of the two interacting qubits model and discuss its thermalization process. After this introductions, we will be able to analyse our results in Sec. 4.3.

4.1 Quantum Coherence and Entanglement

One of the basic properties of quantum systems is the possibility of being in a superposition state, for example $|\psi\rangle = a|0\rangle + b|1\rangle$, where $|0\rangle = (1,0)^T$ and $|1\rangle = (0,1)^T$ form the computational basis. In the density matrix form, this state becomes

$$\rho = \begin{bmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{bmatrix}, \tag{4.1}$$

where the off-diagonal elements represent the coherences of the state. In other words, we call coherent states the ones with off-diagonal elements in their density matrices while the incoherent states are represented by diagonal density matrices (44,45). However, coherence is a basis dependent quantity, which implies that we must fix one basis to define it. Usually this reference basis is selected according to the physical problem, for example, the energy eigenbasis is the most used in studies of thermodynamic processes. It is important to emphasize that the off-diagonal elements cannot assume arbitrary values, because the positive semi-definite condition of the density matrix must be preserved, which implies that

$$|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj}, \tag{4.2}$$

where the subindexes identifies the density matrix element.

There are different ways to measure the coherence of a state, being the relative entropy of coherence one of them. It is defined as

$$C_R(\rho) = \min_{\sigma \in I} D[\rho||\sigma] = S(\rho_d) - S(\rho), \tag{4.3}$$

that is a minimization over all the set of incoherent states I of the relative entropy between the state we want to measure ρ and the incoherent state σ (44). The last equality is obtained by using that we can separate any density matrix into a diagonal part $\rho_d = \sum_i |i\rangle \langle i| \rho |i\rangle \langle i|$ and a coherent part, which allow us to rewrite the relative entropy as $D[\rho||\sigma] = S(\rho_d) - S(\rho) + D[\rho_d||\sigma]$. The minimization over σ of this expression, means that $\rho_d = \sigma$, resulting the last equality in Eq. (4.3). Using this measure, we can prove that the maximally coherent state have maximum values of the off-diagonal elements, resulting in the pure state $|\psi_{MCS}\rangle = d^{-1} \sum_{i=0}^{d} |i\rangle$ that has $C_R(\rho) = \log d$ (45).

Coherence have non-trivial consequences in thermodynamic processes and, for this reason, investigating its effects in the heating-cooling asymmetry can be interesting. One example of how coherence affects thermodynamics, is that coherence in the energy basis can increase the ergotropy of the state, which means it can be exploited as a resource for work extraction (46). However, there are cases where coherence can be detrimental for physical processes, such as in some stages of quantum batteries operations (21) and in thermal relaxation processes (8).

When we have two of more quantum systems, they can share information through classical and quantum correlations. There are a variety of quantum correlations that can be organised as a hierarchy (47), being coherence the necessary condition for all of them. In this work, we will only study the bipartite entanglement (that is the entanglement between two parts). Consider a composite system whose global state can be separated into two subsystems, such as $|\psi\rangle_{AB} = |\alpha\rangle_A \otimes |\beta\rangle_B$, meaning the subsystems are not entangled. On the other hand, when we cannot obtain the global state by any combination of local states $|\alpha\rangle_A$ and $|\beta\rangle_B$, the global state $|\psi\rangle_{AB}$ is said to be entangled (47). In the density matrices formulation, separable states can be decomposed into local pure mixed states,

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B. \tag{4.4}$$

As a consequence, the state is entangled if the decomposition in Eq. (4.4) is not possible (47).

A usual measure of correlations is the mutual information defined as

$$\mathcal{M}(\rho_{AB}) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}), \tag{4.5}$$

where $S(\rho)$ is the von Neumann entropy, ρ_A and ρ_B are parts of the global state ρ_{AB} (33). However, this quantity indicates how much information is shared through both classical and quantum correlations, meaning it cannot be used alone as a signature of quantumness. Because, in this work, we will only use bipartite entanglement, we can use a more specific quantifier. The entanglement of formation (EoF), that is defined as

$$E_f(\rho) = H\left(\frac{1 + \sqrt{1 - C(\rho)^2}}{2}\right), \tag{4.6}$$

where $H(x) = -x\log_2(x) - (1-x)\log_2(1-x)$ is the Shannon entropy and

$$C(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \tag{4.7}$$

is the concurrence for the case of two qubits*, where $\lambda_i > \lambda_j$ for $i < j$ are the square root of the eigenvalues of the matrix $R = \rho\tilde{\rho}$. To find R, first we take complex conjugation of each matrix element obtaining ρ^* and then we compute the time-reversed matrix $\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$ (48).

Entanglement is widely used as quantum resource in communication and computation (49), and it can also bring advantages in thermodynamics processes. For example, refrigerators can make entangled systems reach lower temperatures than no-correlated states (50). Another example, is that entanglement can help extract more work in quantum batteries (37).

There are other methods to quantify entanglement and coherence in quantum systems, but the presented discussion is enough for this work. In the next section, we will explain the thermal relaxation of the two interacting qubits model and will present our results in Sec. 4.3.

4.2 Thermal Relaxation of Two Interacting Qubits

The two interaction qubits model we will use can be described by

$$H_S = -\frac{\omega_0}{2}\sigma_1^z - \frac{\omega_0}{2}\sigma_2^z + J[\sigma_1^+\sigma_2^- + \sigma_1^-\sigma_2^+], \tag{4.8}$$

where $\sigma_i^+ = |1\rangle\langle 0|$, $\sigma_i^- = |0\rangle\langle 1|$, and σ_i^z is the Pauli matrix in z acting on qubit i. We will consider that both qubits have the same energy gap $\frac{\omega_0}{2}$. The qubits can exchange energy and information through the XX-interaction model with strength J $(48, 51)$. Depending on the value of J, the final state of the dynamics can have more coherence and even stronger correlations, such as entanglement, which will be discussed later in this section.

To study the heating-cooling asymmetry with this model, we will consider that one qubit is initially at a colder temperature T_c and the other qubit is initially at a hotter temperature T_h. Because the bath is at an intermediate temperature T_w, the colder qubit will heat up and the hotter qubit will cool down. To genuinely study heating and cooling, it is important to assure each qubit is at a thermal state so we can define a physical temperature to it. If we add off-diagonal elements in each qubit's density matrix, that is, coherence, they would not be thermal states and the terms "heating" and "cooling" become inaccurate. Therefore, to study the effects of coherence in heating-cooling asymmetry with this model, we will use an initial X-state

$$\rho = \rho_c \otimes \rho_h + \chi, \tag{4.9}$$

* The general definition can be found in Ref. (47)

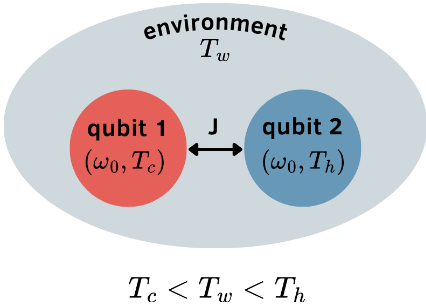

Figure 2 – Illustration of two interacting qubits in contact wit a global bath. The red qubit is at a colder temperature T_c and will heat up to the bath's temperature T_w. The blue qubit is at a hotter temperature T_h and will cool down to the bath's temperature T_w. The qubits can interact with a strength J.

where ρ_c and ρ_h are the thermal states of the cold and hot qubits, respectively, and χ is a matrix defined in computational basis as

$$\chi = e\left|01\right\rangle\left\langle 10\right| + e^*\left|10\right\rangle\left\langle 01\right| + f\left|00\right\rangle\left\langle 11\right| + f^*\left|11\right\rangle\left\langle 00\right|, \tag{4.10}$$

which means it contains the coherences of the global state. These coherences only exist for the global state because $\mathrm{Tr}_c[\chi] = \mathrm{Tr}_h[\chi] = 0$. This particular format is important because it allow us to include coherences while keeping thermal local states for each qubit, which means that $\mathrm{Tr}_c[\rho] = \rho_h$ and $\mathrm{Tr}_h[\rho] = \rho_c$.

As discussed in Sec. 4.1, the positivity of every density matrix imposes the constraint shown in Eq. (4.2) on the off-diagonal elements. In our case, the matrix form of the initial density matrix of our state is

$$\rho = \begin{bmatrix} \frac{e^{\omega_0(\beta_c+\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} & 0 & 0 & f \\ 0 & \frac{e^{\omega_0(\beta_c-\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} & e & 0 \\ 0 & e^* & \frac{e^{-\omega_0(\beta_c-\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} & 0 \\ f^* & 0 & 0 & \frac{e^{-\omega_0(\beta_c+\beta_h)/2}}{\mathcal{Z}_c\mathcal{Z}_h} \end{bmatrix}, \tag{4.11}$$

where $\beta_i = 1/T_i$ and the partition function is $\mathcal{Z}_i = 2\cosh(\beta_i\omega_0/2)$ with $i = (c, h)$. After its diagonalization, we obtain that the maximum value for $|e|$ and $|f|$ is $1/\mathcal{Z}_c\mathcal{Z}_h$. Even though this initial state can have coherence, this limitation for the off-diagonal elements makes impossible to have entanglement according to the requirements in Ref. (52).

To study the thermal relaxation of this model and initial state, we will use the Davies maps presented in Sec. 2.2.1. As we previously discussed, the steady state of this dynamics is always a thermal state, which enable us to find that

$$
\begin{aligned}
\rho_{ss}^{G} &= \frac{e^{-\beta_w H_S}}{\mathcal{Z}_S} \\
&= \frac{1}{\mathcal{Z}_S}\begin{bmatrix} e^{\beta_w\omega_0} & 0 & 0 & 0 \\ 0 & e^{\beta_w J} & 0 & 0 \\ 0 & 0 & e^{-\beta_w J} & 0 \\ 0 & 0 & 0 & e^{-\beta_w\omega_0} \end{bmatrix} \\
&= \frac{1}{\mathcal{Z}_S}\begin{bmatrix} e^{\beta_w\omega_0} & 0 & 0 & 0 \\ 0 & \cosh(\beta_w\omega_0) & -\sinh(\beta_w J) & 0 \\ 0 & -\sinh(\beta_w J) & \cosh(\beta_w\omega_0) & 0 \\ 0 & 0 & 0 & e^{-\beta_w\omega_0} \end{bmatrix},
\end{aligned}
\tag{4.12}
$$

where $\mathcal{Z}_S = 2(\cosh(\beta_w\omega_0) + \cosh(\beta_w J))$ is the partition function with $\beta_w = 1/T_w$ and H_S being the Hamiltonian from Eq. (4.8). The second equality is in the energy basis and the last equality is in the computational basis. Now that we know the global steady state, we go back to the discussion about the emergence of entanglement due to the interaction of the qubits. We can use entanglement of formation (defined in Eq. (4.6)) to identify which parameter we can obtain entanglement between the qubits. First, we need to compute the concurrence (defined in Eq. (4.7)) for this state. In Appx. A, we find that the square root of the eigenvalues of R (defined in Sec. 4.1) are in decreasing order $\lambda_1 = e^{\beta_w J}/\mathcal{Z}_S$, $\lambda_2 = \lambda_3 = 1/\mathcal{Z}_S$, and $\lambda_4 = e^{-\beta_w J}/\mathcal{Z}_S$. Fig. 3 show that concurrence increase for $J > 0.8814$, resulting in an increase of the entanglement of formation. So we conclude that this model can have entanglement for stronger interaction between the qubits. Additionally, it is also possible to compute the mutual information defined in Eq.

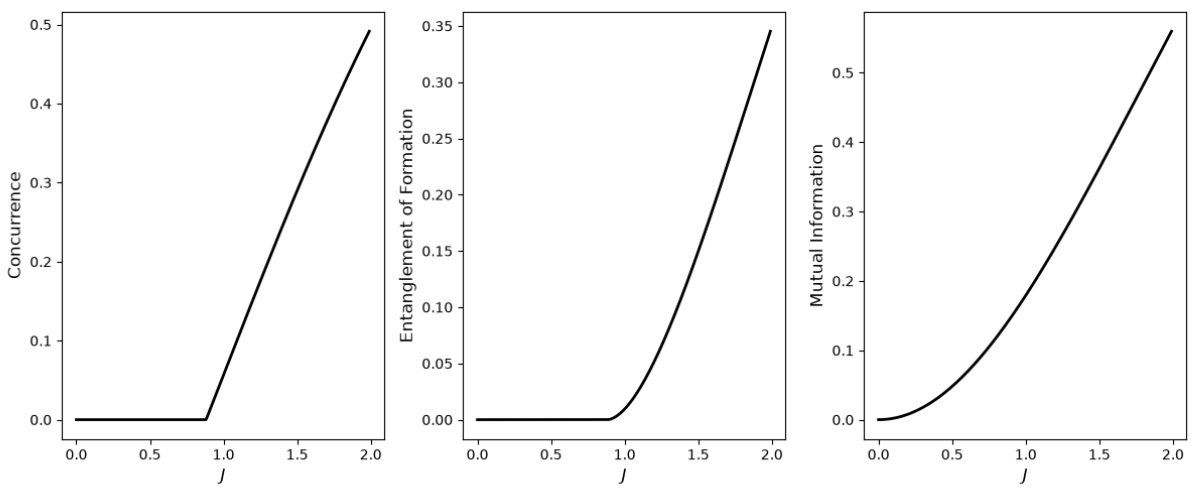

Figure 3 – Quantifiers of correlations in the global thermal state (Eq. (4.12)) caused by the qubits interaction. Concurrence (left), entanglement of formation (right), and mutual information as a function of the interaction parameter J.

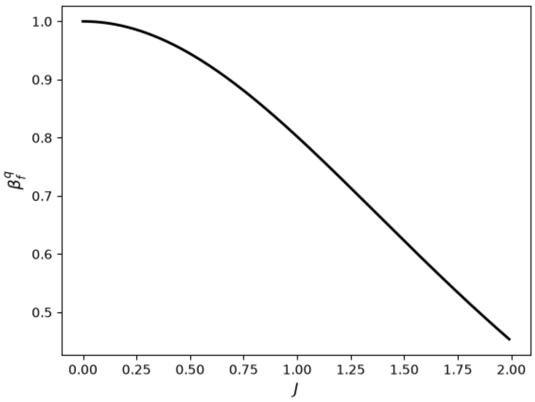

Figure 4 – Final temperature of the qubits (Eq. (4.14)) as a function of the interaction parameter J.

(4.5) and shown the right plot of Fig. 3. We see that the mutual information increase for higher J, but it is bigger than zero before entanglement of formation, which means that other types of correlations emerge even for weak interaction between the qubits.

However, the appearance of correlation between the qubits affects their final states. When we trace out one of the qubits from the global steady state (Eq. (4.12)), we find the local steady state

$$\rho_{ss}^q = \frac{1}{\mathcal{Z}_q}\begin{bmatrix} e^{\beta_w \omega_0} + \cosh(\beta_w J) & 0 \\ 0 & e^{-\beta_w \omega_0} + \cosh(\beta_w J) \end{bmatrix}, \tag{4.13}$$

where $\mathcal{Z}_q = 2\cosh(\beta_f \omega_0) + 2\cosh(\beta_f J)$. Because in a global bath both qubits equilibrate to the same local temperature, it makes no difference which one we trace out. From the final state of the qubit, we can compute its final temperature, obtaining

$$\beta_f^q = \frac{1}{\omega_0}\ln\left(\frac{e^{\beta_w \omega_0} + \cosh(\beta_w J)}{e^{-\beta_w \omega_0} + \cosh(\beta_w J)}\right), \tag{4.14}$$

which is smaller than the bath's temperature for $J > 0$, as shown in Fig. 4. It must be emphasised that the global system thermalises to the bath's temperature, as we calculated in Eq. (4.12). However, because the interaction between the qubits generates correlations (as signalised by the mutual information), tracing out one of them erases part of the information, which changes the qubit's populations resulting in a different final temperature.

Now that we discussed the interaction model between the qubits and the structure of the initial and final states, we can analyse how the heating-cooling asymmetry behaves for our system. In the next section, we present our results and how we obtained them.

4.3 Asymmetry for Two Interacting Qubits

Before solving the dynamics for our model and analyse the asymmetry, it is important to guarantee we are making a fair comparison between heating and cooling. In this sense, we must choose initial states for the qubits that are thermodynamically equivalent, which is not the same as having equidistant initial temperatures from equilibrium temperature. Given that the asymmetry only appears for far-from-equilibrium systems, the criteria we will use for thermodynamic equivalence will be the nonequilibrium free energy defined by Eq. (3.8). As we discussed in Sec. 3.1, states with the same nonequilibrium free energy also have the same relative entropy from equilibrium. So, we choose a colder (ρ_c) and hotter (ρ_c) initial states for the qubits that have the same relative entropy from their final local steady state (ρ^q_{ss}),

$$D[\rho_c||\rho^q_{ss}] = D[\rho_h||\rho^q_{ss}] = D_0, \tag{4.15}$$

where we use the relative entropy definition from Eq. (3.6). This equality can be numerically solved by varying the initial temperature of the qubit's thermal state and computing its relative entropy from ρ^q_{ss}. This will result is the orange curve in Fig. 5, where the orange dot indicates the equilibrium state with inverse temperature β^q_f. Then, we can find the points where this orange curve is equal to the initial relative entropy D_0 (black line). Using that the population of the ground state ($\langle 0|\,\rho\,|0\rangle = \exp(\omega_0/2T_i)/2\cosh(\omega_0/2T_i)$ with $i = (c, h)$), we can find the initial cold (blue dot) and hot (red dot) temperatures that the qubits must have to be thermodynamically equivalent.

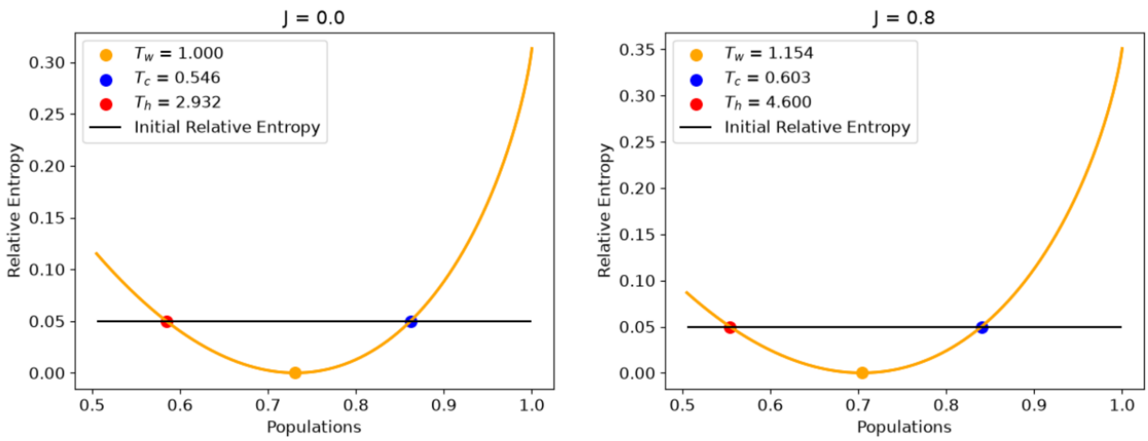

Figure 5 – Relative entropy of a thermal qubit and its equilibrium state as a function of its ground state population. The orange curve is obtained by varying the qubit's temperature and the orange dot corresponds to the equilibrium state. The black line indicates the chosen initial relative entropy that both hot (red dot) and cold (blue dot) qubits must have. The left plot is the no-interaction case, while the right plot show the interaction case with $J = 0.8$.

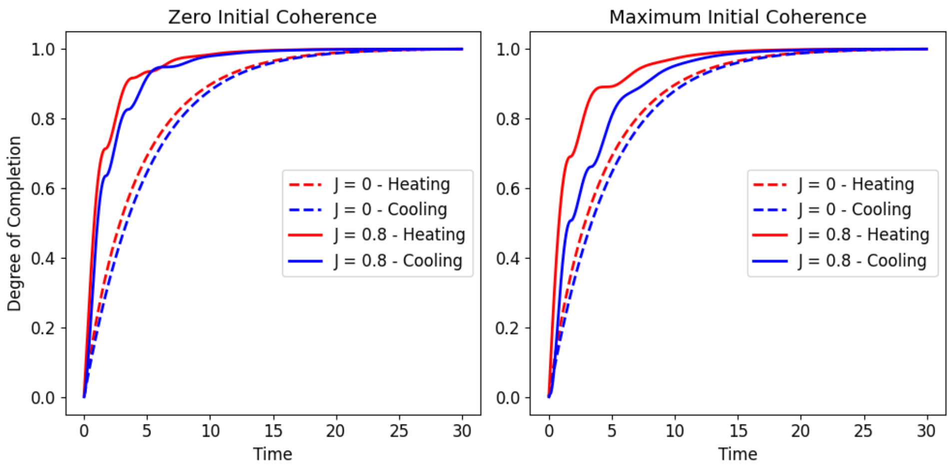

Figure 6 – Degree of completion for XX-model. Each line correspond to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

With this initialization of our system, we can analyse the heating and cooling asymmetry. In the next figures, heating and cooling are represented in red and blue, respectively. Dashed lines indicate the non-interaction case ($J = 0$), while solid lines represent the interaction case without entanglement ($J = 0.8$). The degree of completion (defined by Eq. (3.24)) is shown in Fig. 6 for the case without initial coherence (left) and the case with maximum initial coherence (right). When comparing heating and cooling, the red lines are above their corresponding blue lines, so we conclude that heating is faster than cooling. Using this same idea, we see that the interaction between qubits allows a faster relaxation. Furthermore, only for the interaction case, the initial coherence seems to affect the intermediary dynamics by delaying cooling, which increases the asymmetry while keeping the total relaxation time unchanged. These same observations can be also obtained from the statistical velocity (defined by Eq. (3.22)) shown in Fig. 7. In addition to this, note that the interaction case have many oscillations in both degree of completion and statistical velocity, which will be discussed later.

Previous works have stated that heating is faster than cooling because it has higher entropy production (10, 18). As we noticed that the interaction case relaxate faster than the no-interaction case, we could try to also use the argument that the faster process has higher entropy production. In Fig. 8, we show the entropy production rate (defined in Eq. (3.5)) as a function of time. First, heating without interaction have higher entropy production than cooling which agrees with previous works (10, 18). However, the interaction case have many oscillations and heating entropy production is not always bigger than cooling entropy production. Additionally, the entropy production of the interaction case becomes negative for some time intervals. If we recall that entropy is related to the lack of information, negative entropy production means the system is gaining informa-

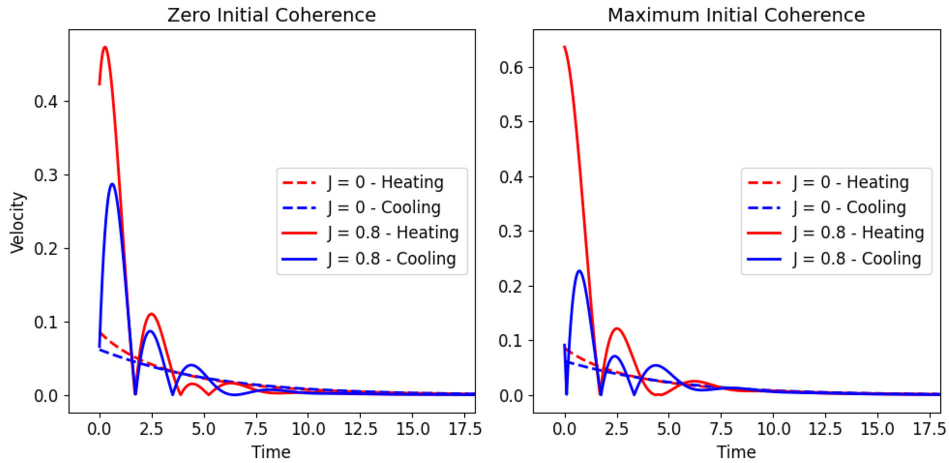

Figure 7 – Statistical velocity for XX-model. Each line correspond to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

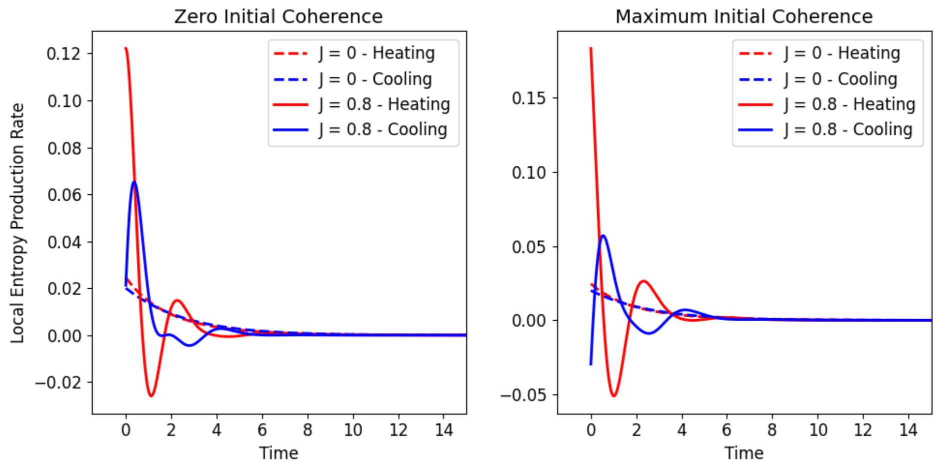

Figure 8 – Entropy production rate for the XX-model. Each line corresponds to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

tion instead of losing to the environment, which seems like a violation of the second law of thermodynamics. This gain of information is also called information backflow and is related with non-Markovian dynamics (53). However, we are using a global Markovian master equation as defined in Chap. 2. A possible explanation to this dichotomy is that, by limiting our observation to a subpart of the system, we also ignore correlations with the other part, which can seem like memory effects (54). This problem have already been studied before and it was shown that negative entropy production can be a witness of non-Markovianity (54, 55). So a conclusion from this analysis is that non-Markovianity may help the systems to have faster relaxation.

Until this point, we have only discussed an interaction case that do not generate

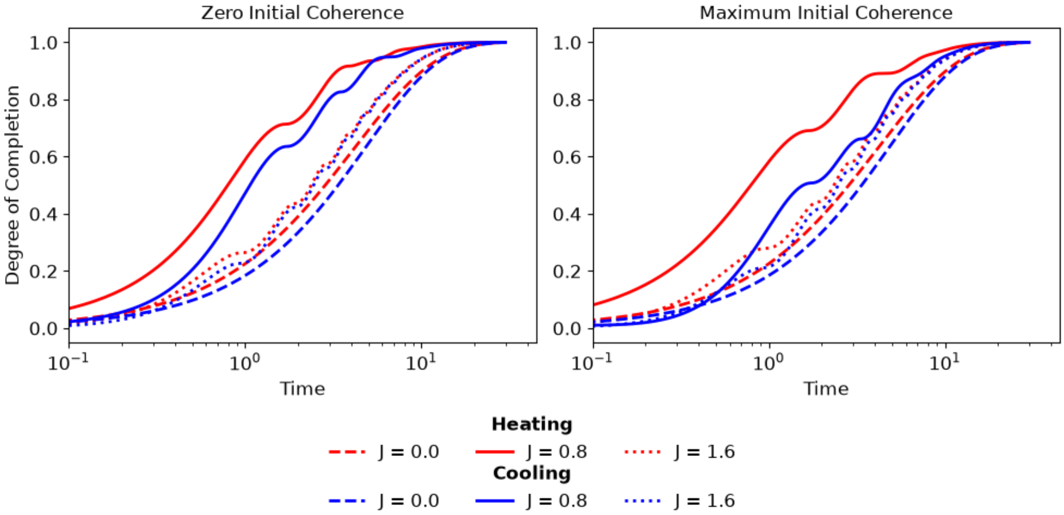

Figure 9 – Degree of completion for heating (red) and cooling (blue) states with zero (left) or maximum (right) initial coherence. The interaction parameters used are $J = 0$ (dashed), $J = 0.8$ (solid), and $J = 1.6$ (dotted).

entanglement between the qubits. According to Fig. 3, we can increase J to obtain a final entangled state for the qubits. Choosing $J = 1.6$, Fig. 9 show that the presence of entanglement (dotted lines) reduces the heating-cooling asymmetry making the curves closer. Again the initial coherence do not change the overall thermalisation for $J = 1.6$. When compared to our previous interaction case without entanglement (solid lines), we notice that the thermal relaxation becomes slower, but it is still faster than the non-interacting case (dashed lines). Therefore, the relation between the acceleration of thermal relaxation and the increase of J it is not straightforward.

5 HEATING-COOLING ASYMMETRY: CONTINUOUS-VARIABLE CASE

Continuous-variable (CV) systems are described by observables with continuous spectra, typically associated with infinite-dimensional Hilbert spaces. An important subclass of CV systems are Gaussian states and operations, as they have been widely applied in communication protocols, quantum field theory, solid state physics, and open quantum systems (47). In Sec. 5.1, we present the basic theory of Gaussian systems and the main references for this section are (47,56). In this work, we will investigate the heating-cooling asymmetry for the three cases of Gaussian states: thermal states (Sec. 5.3), displaced states (Sec. 5.4), and squeezed states (Sec. 5.5). As before, we will use the thermal kinematics approach to understand the asymmetry, however, we will use the Wigner-Fisher information instead of the quantum fisher information, which will result in some small changes on the thermal kinematics variables, which will be discussed in Sec. 5.2

5.1 Gaussian States

An important example of CV systems is the quantised electromagnetic field, whose modes behave as independent harmonic oscillators. Despite being described by continuous variables, the energy of each mode is quantised in equally spaced levels, with transitions occurring through the absorption or emission of energy quanta known as photons. For simplicity, let us consider an electromagnetic field inside an optical cavity, where only N relevant modes are taken into account. In this case, the Hamiltonian can be written as the sum of independent modes, $H = \sum_{k=1}^{N} H_k$, with each mode described by the harmonic oscillator Hamiltonian,

$$H_k = \hbar\omega_k \left(a_k^\dagger a_k + \frac{1}{2}\right). \tag{5.1}$$

Here, $a_k^\dagger$ and a_k are the creation and annihilation operators of the mode k and satisfy the commutation relations $[a_i, a_j^\dagger] = \delta_{ij}$ and $[a_i^\dagger, a_j^\dagger] = [a_i, a_j] = 0$. These operators have these names because they create or annihilate a photon with wave vector $\mathbf{k}$, changing the mode energy by one quantum. The quantum harmonic oscillator can equivalently be described in terms of the position and momentum operators, also called quadratures,

$$q_k = \frac{1}{\sqrt{2}}(a_k^\dagger + a_k), \tag{5.2}$$

$$p_k = \frac{i}{\sqrt{2}}(a_k^\dagger - a_k). \tag{5.3}$$

For N-mode systems, the quadratures can be grouped into a vector $R = (R_1, \ldots, R_N)^T = (q_1, p_1, ..., q_N, p_N)^T$, allowing the commutation relations to be condensed in $[R_i, R_j] = i\Omega_{ij}$, where

$$\Omega = \bigoplus_{k=1}^{N} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \tag{5.4}$$

is the N-mode symplectic form. The quadratures operators define the phase space $\mathcal{P} = (\mathbb{R}^{2N}, \Omega)$, a real 2N-dimensional space provided with the symplectic structure Ω. As a consequence, quantum states are represented as phase-space regions instead of single points like in classical mechanics.

The eigenstates of each Hamiltonian H_k in Eq. (5.1) are the Fock states $\left|n\right\rangle_k$, where n denotes the number of photons in mode k. As a consequence, the basis for the full multimode Hamiltonian is given by the tensor product $\left|n\right\rangle_{\mathbf{k}} = \left|n\right\rangle_1 \otimes \left|n\right\rangle_2 \otimes \cdots \otimes \left|n\right\rangle_N$. Each Fock basis is bounded from below by the vacuum state $\left|0\right\rangle_k$, defined by $a_k \left|0\right\rangle_k = 0$. Although the Fock representation is important for describing quantised fields, it becomes less convenient when dealing with states containing a large number of photons, such as laser fields. In these cases, an alternative representation is to use coherent states. A single-mode coherent state can be written as a linear combination of Fock states,

$$\left|\alpha\right\rangle_k = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \left|n\right\rangle_k. \tag{5.5}$$

Coherent states can also be described as the application of the Weyl operator, also known as the displacement operator, to the vacuum state,

$$\left|\alpha\right\rangle_k = D_k(\alpha) \left|0\right\rangle_k, \tag{5.6}$$

where the Weyl operator is

$$D_k(\alpha) = e^{\alpha a_k^\dagger - \alpha^* a_k}, \tag{5.7}$$

with $\alpha \in \mathbb{C}$ being the displacement parameter. For the N-mode case, the displacement operator can be written in terms of the canonical operator vector R as

$$D(\mathbf{r}) = e^{R^T \Omega \mathbf{r}}, \tag{5.8}$$

where $\mathbf{r} = (q'_1, p'_1, \ldots, q'_N, p'_N)$ is a vector in phase-space $\mathcal{P}$ specifying the displacement applied for each quadrature.

Until this moment, we have been using density matrices to describe our states. However, the density matrices of CV systems are infinite-dimensional, making this approach mathematically complicated. An alternative is to work with equivalent phase-space representations. The characteristic function is a multivariable function obtained through the Fourier-Weyl transform of the density operator (56), being written as

$$\chi_\rho(\mathbf{r}) = \mathrm{Tr}[\rho D(\mathbf{r})]. \tag{5.9}$$

In this work, we will consider only the symmetric-ordered characteristic function defined above; other orderings can be found in Refs. (56, 57). Although χ_ρ fully characterises a quantum state, it is still expressed in terms of displacement variables rather than quadrature operators. The full transition to phase-space can be done by taking the Fourier transform of the characteristic function, resulting in the Wigner quasiprobability distribution,

$$W_\rho(\mathbf{q}, \mathbf{p}) = \frac{1}{\pi^N} \int_{\mathbb{R}^N} \langle \mathbf{q} + \mathbf{x} | \, \rho \, | \mathbf{q} - \mathbf{q}' \rangle \, e^{2i\mathbf{x}.\mathbf{p}} d^N \mathbf{x}. \tag{5.10}$$

Here, $\mathbf{q}$ and $\mathbf{p}$ are N-dimensional vectors indicating the coordinates of a point in the phase-space. The vector $\mathbf{x}$ is a symmetric displacement around $\mathbf{q}$, being the integration variable and $q_k\mathbf{x} = q_k\mathbf{x}$. The Wigner distribution is called a quasiprobability because, unlike classical probability distributions, it may take negative values. Still, its marginalisation reproduces the correct probability distribution associated with the quadrature observables (57).

Among the different classes of continuous-variable states, Gaussian states play an important role due to their mathematical simplicity and experimental relevance. As the name suggests, the Wigner distribution associated with these states is a Gaussian function in phase space. For a single variable, a Gaussian function can be written as

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right), \tag{5.11}$$

where $\mu = \langle x \rangle$ is the mean value (first moment) and $\sigma^2 = \langle (x - \langle x \rangle)^2 \rangle$ is the variance (second moment). These two quantities fully characterise a Gaussian distribution. Similarly, multimode Gaussian states are fully characterised by the first and second moments of the quadrature operators. The first moments are grouped into the mean vector $\mathbf{v}$, whose elements are

$$v_j = \langle R_j \rangle_\rho \,, \tag{5.12}$$

while the second moments are written in the covariance matrix Θ, with elements

$$\Theta_{ij} = \frac{1}{2} \langle R_i R_j + R_j R_i \rangle_\rho - \langle R_i \rangle_\rho \langle R_j \rangle_\rho \,, \tag{5.13}$$

where $\langle O \rangle_\rho \equiv \mathrm{Tr}[\rho O]$. With these definitions, the Wigner distribution becomes

$$W(\mathbf{x}) = \frac{1}{\pi^N \sqrt{|\Theta|}} \exp\left(-(\mathbf{x} - \mathbf{v})^T \Theta^{-1} (\mathbf{x} - \mathbf{v}) \right), \tag{5.14}$$

where $\mathbf{x} \in \mathbb{R}^{2N}$ and $|\Theta| = \det(\Theta)$. Important examples of Gaussian states are coherent states, thermal states, and the vacuum state.

In addition to characterising Gaussian states, we are interested in the operations that act upon them. Although general quantum operations may destroy Gaussianity,

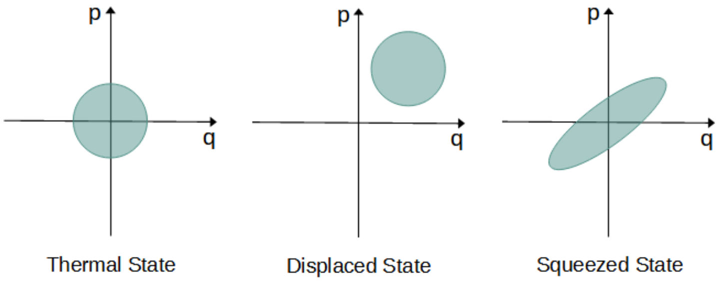

Figure 10 – Illustration of a single-mode thermal state, displaced coherent state, and squeezed coherent state in the phase space.

Gaussian operations preserve the Gaussian form of the state in phase space. In this work, we will only analyse single-mode Gaussian states, where the relevant unitary Gaussian operations are displacement operations, defined in Eq. (5.7), and squeezing operations. The displacement operator, as the name suggests, shifts the whole Gaussian state without deforming the distribution. In other terms, it varies the mean vector while keeping the covariance matrix unchanged. On the other hand, the squeezing operator for a single mode is defined as

$$S_k(z) = e^{-\frac{1}{2}(z^* a_k - z a_k^\dagger)}, \tag{5.15}$$

where $z = re^{i\phi_s}$ with r being the squeezing parameter and ϕ_s being the squeezing phase responsible to rotate the state. The squeezing operator deforms the state in the phase space by compressing one quadrature. As a consequence of the uncertainty principle, the other quadrature will expand. Mathematically, this means that squeezing operations only affect the covariance matrix, preserving the mean vector. An illustration of how these operations affect a Gaussian state is shown in Figure 10. We will not use the multimode squeezing operator in this work, but its definitions can be found in Refs. (56, 57).

5.2 Thermal Relaxation of Gaussian States

As discussed in Chap. 2, the dynamics of an open system can be described by the Lindblad master equation. In our case, we will consider a single bosonic mode (Eq. (5.1) with $k = 1$) interacting with a thermal bath and evolving in time according to Davies maps, which has already been calculated in the example in Sec. 2.2.1.2,

$$\frac{d}{dt}\rho = -i[H, \rho] + \Gamma(\bar{n} + 1)\mathcal{D}_a(\rho) + \Gamma\bar{n}\mathcal{D}_{a^\dagger}(\rho), \tag{5.16}$$

where Γ is the dissipation rate and $\bar{n} = 1/(e^{\beta\omega} - 1)$ is the Bose distribution. Recalling the jump operators of Eq. (2.50), we can rewrite $\mathcal{D}^-(\rho) = \Gamma(\bar{n} + 1)\mathcal{D}_a(\rho)$ and $\mathcal{D}^+(\rho) = \Gamma\bar{n}\mathcal{D}_{a^\dagger}(\rho)$. Because this is a Davies maps, we know that the final state will always be a thermal state with mean vector $\mathbf{v}_{eq} = 0$ and a covariance matrix equal to $\Theta_{eq} = f(\beta)\mathbb{I}$, where $f(\beta) = \bar{n} + 1/2$.

As elaborated in Sec. 5.1, it is convenient to use the phase-space representation instead of the density matrix. For this reason, we will rewrite Eq. (5.16) in terms of the mean vector and the covariance matrix using that $\frac{d}{dt}\langle O\rangle_\rho = \mathrm{Tr}[O\frac{d}{dt}\rho]$. The mean vector of a single mode has the form $\mathbf{v}_t = (\langle a\rangle, \langle a^\dagger\rangle)^T$. To find its time derivative, we compute

$$\frac{d}{dt}\langle a\rangle = -\left(i\omega + \frac{\Gamma}{2}\right)\langle a\rangle \tag{5.17}$$

$$\frac{d}{dt}\langle a^\dagger\rangle = \left(i\omega - \frac{\Gamma}{2}\right)\langle a^\dagger\rangle, \tag{5.18}$$

which gives the differential equation

$$\dot{\mathbf{v}}_t = \Lambda\mathbf{v}_t, \tag{5.19}$$

where

$$\Lambda = -\frac{1}{2}\begin{bmatrix}\Gamma + 2i\omega & 0 \\ 0 & \Gamma - 2i\omega\end{bmatrix}. \tag{5.20}$$

The covariance matrix of a single mode is structured as

$$\Theta_t = \begin{bmatrix}\frac{1}{2}(\langle a^\dagger a\rangle + \langle aa^\dagger\rangle) - \langle a^\dagger\rangle\langle a\rangle & \langle a^2\rangle - \langle a\rangle^2 \\ \langle a^{\dagger 2}\rangle - \langle a^\dagger\rangle^2 & \frac{1}{2}(\langle a^\dagger a\rangle + \langle aa^\dagger\rangle) - \langle a^\dagger\rangle\langle a\rangle\end{bmatrix}. \tag{5.21}$$

To take its time derivative, we first compute

$$\frac{d}{dt}\langle aa^\dagger\rangle = -\Gamma(\langle aa^\dagger\rangle - \bar{n} - 1), \tag{5.22}$$

$$\frac{d}{dt}\langle a^\dagger a\rangle = -\Gamma(\langle aa^\dagger\rangle - \bar{n}), \tag{5.23}$$

$$\frac{d}{dt}\langle a^2\rangle = -(2i\omega + \Gamma)\langle a^2\rangle, \tag{5.24}$$

$$\frac{d}{dt}\langle a^{\dagger 2}\rangle = (2i\omega - \Gamma)\langle a^{\dagger 2}\rangle, \tag{5.25}$$

to obtain the following differential equation called the Lyapunov Equation,

$$\dot{\Theta}_t = \Lambda\Theta + \Theta\Lambda^\dagger + F, \tag{5.26}$$

where $F = \Gamma f(\beta)\mathbb{I}$. Therefore, by solving Eqs. (5.19) and (5.26), we find that the time evolution of a single-mode Gaussian state is given by

$$\mathbf{v}_t = e^{-\Gamma t}(\langle a\rangle_0 e^{-i\omega t}, \langle a^\dagger\rangle_0 e^{i\omega t})^T \tag{5.27}$$

$$\Theta_t^{11} = \Theta_t^{22} = c_0 e^{-\Gamma t} + f(\beta), \tag{5.28}$$

$$\Theta_t^{12} = (\Theta_t^{21})^* = d_0 e^{-(\Gamma + 2i\omega)t}, \tag{5.29}$$

where c_0, d_0, $\langle a\rangle_0$, and $\langle a^\dagger\rangle_0$ are constants obtained from the initial conditions.

To analyse the thermal relaxation of a Gaussian state from a thermodynamic perspective, we will rewrite some information and thermodynamic quantities we defined in Chap. 3 in terms of the mean vector and covariance matrix. Because the asymmetry only occurs in the far-from-equilibrium regime, it will be useful to find the nonequilibrium free energy of a Gaussian state. As we did in Sec. 3.1, we will start with the average energy of a Gaussian state with Wigner distribution $W(\mathbf{x})$, which is defined as

$$E(W(\mathbf{x})) = \omega \int |\mathbf{x}|^2 W(\mathbf{x}) d^2\mathbf{x}. \tag{5.30}$$

Using that the Wigner distribution of an arbitrary thermal state is

$$W_{th}(\mathbf{x}) = \frac{e^{-|\mathbf{x}|^2/f(\beta)}}{\pi f(\beta)}, \tag{5.31}$$

we can substitute $|\mathbf{x}|^2 = -f(\beta)(\ln(\pi f(\beta)) + \ln W_{th}(\mathbf{x}))$ in the average energy to obtain

$$\begin{aligned} E(W(\mathbf{x})) &= -\omega f(\beta) \int W(\mathbf{x})(\ln(\pi f(\beta)) + \ln W_{th}(\mathbf{x}))d^2\mathbf{x} \\ &= -\omega f(\beta) \left[\ln(\pi f(\beta)) + \int W(\mathbf{x}) \ln W_{th}(\mathbf{x}) d^2\mathbf{x}\right], \end{aligned} \tag{5.32}$$

where we used $\int W(\mathbf{x})d^2\mathbf{x} = 1$. Similar to what we did in Eq. 3.7, we will introduce the Wigner entropy defined as

$$S_W(W(\mathbf{x})) = -\int W(\mathbf{x}) \ln W(\mathbf{x}) d^2\mathbf{x} = 1 + \ln \pi + \frac{1}{2} \ln |\Theta|, \tag{5.33}$$

and find

$$\begin{aligned} E(W(\mathbf{x})) &= -\omega f(\beta) \left[\ln(\pi f(\beta)) + \int W(\mathbf{x}) \ln W_{th}(\mathbf{x}) d^2\mathbf{x} + S_W(W(\mathbf{x})) - S_W(W(\mathbf{x}))\right] \\ &= -\omega f(\beta) \left[\ln(\pi f(\beta)) + \int W(\mathbf{x}) \ln W_{th}(\mathbf{x}) d^2\mathbf{x} - \int W(\mathbf{x}) \ln W(\mathbf{x}) d^2\mathbf{x} - S_W(W(\mathbf{x}))\right] \end{aligned} \tag{5.34}$$

where we can identify the Wigner relative entropy, that is defined as

$$\begin{aligned} K[W_1(\mathbf{x})||W_2(\mathbf{x})] &= \int W_1(\mathbf{x}) \ln \frac{W_1(\mathbf{x})}{W_2(\mathbf{x})} d^2\mathbf{x} \\ &= -1 + \frac{1}{2}\left[(\mathbf{v}_1 - \mathbf{v}_2)^\dagger \Theta_2^{-1} (\mathbf{v}_1 - \mathbf{v}_2) + \mathrm{Tr}(\Theta_2^{-1}\Theta_1) + \ln \frac{|\Theta_2|}{|\Theta_1|}\right]. \end{aligned} \tag{5.35}$$

As a result, the internal energy of a Gaussian system is

$$E(W(\mathbf{x})) = \omega f(\beta) K[W(\mathbf{x})||W_{th}(\mathbf{x})] + \omega f(\beta) S(W(\mathbf{x})) + F_{\text{eq}}, \tag{5.36}$$

where $F_{\text{eq}} = -\omega f(\beta) \ln(\pi f(\beta))$ is the equilibrium free energy of a thermal state. Given that for Gaussian states the Wigner entropy satisfy conditions analogous to the von

Neumann entropy (47), we can extend the Helmholtz free energy to $F(W) = E(W) - \omega f(\beta) S_W(W)$. Using this free energy and considering that $W_{th}(\mathbf{x}) = W_{\text{eq}}(\mathbf{x})$ is the equilibrium state, we can define a nonequilibrium free energy of a Gaussian state as

$$F_{\text{neq}}(W(\mathbf{x})) = \omega f(\beta) K[W(\mathbf{x})||W_{\text{eq}}(\mathbf{x})] + F_{\text{eq}}, \tag{5.37}$$

which is similar to Eq. 3.8. So, the Wigner relative entropy (Eq. (5.35)) is an indicator of how far from the equilibrium our system is. For this reason, when studying the asymmetry in Gaussian states, we will also impose that the hot and cold initial states have the same initial Wigner relative entropy.

As discussed in Chap. 3, another approach to study the nonequilibrium regime is using ergotropy defined in Eq. (3.10). If we substitute (5.36) into Eq. (3.10), we find

$$\mathcal{E}(W(\mathbf{x})) = \omega f(\beta)(K[W(\mathbf{x})||W_{th}(\mathbf{x})] - K[W_\pi(\mathbf{x})||W_{th}(\mathbf{x})]), \tag{5.38}$$

where $W_\pi(\mathbf{x})$ is the Wigner function of the passive state that has inverse temperature β_π defined by $f(\beta_\pi) = \sqrt{|\Theta|}$. Because the passive state of a Gaussian state is always thermal, we can assume $W_{th} = W_\pi$ such that ergotropy becomes

$$\mathcal{E}(W(\mathbf{x})) = \omega f(\beta_\pi) K[W(\mathbf{x})||W_\pi(\mathbf{x})]. \tag{5.39}$$

From this expression, we find that every squeezed and displaced state have ergotropy meaning they are in the nonequilibrium regime (58). Moreover, squeezing operations can also create coherences in the state, so squeezed states can have quantum resources. An interesting result from Ref. (9) is that squeezed states with higher initial ergotropy relaxate faster than displaced state, what has been named ergotropic Mpemba effect. They also showed that, even when the initial ergotropy is the same, the squeezed state also relaxate faster than the displaced state. These results inspired the study of how displacement and squeezing may affect the thermal relaxation asymmetry.

However, instead of assuming the arbitrary thermal state is equal to the passive state, we can assume it is the equilibrium state ($W_{th} = W_{\text{eq}}$). Then, we make the time derivative on Eq. (5.38) and find that the entropy production for Gaussian states are

$$\begin{aligned}\Pi_W &= -\frac{d}{dt} K[W(\mathbf{x})||W_{th}(\mathbf{x})] \\ &= -\frac{d}{dt} K[W_\pi(\mathbf{x})||W_{th}(\mathbf{x})] - \frac{\dot{\mathcal{E}}}{\omega f(\beta)} \\ &= \Pi_\pi - \frac{\dot{\mathcal{E}}}{\omega f(\beta)},\end{aligned} \tag{5.40}$$

which is similar to Eq. (3.12). So it is also possible to separate the entropy production of Gaussian states into a passive and an ergotropic contribution.

Now that we briefly redefined some thermodynamic quantities for Gaussian states, we will discuss how the thermal kinematics change in this regime. As discussed in Sec. 3.2, Fisher information is the essential quantity for the thermal kinematics approach of heating-cooling asymmetry. To find a Fisher information for Gaussian states, we will follow the same procedure of Sec. 3.2 and make a second order Taylor expansion in a parameter θ of the Wigner relative entropy (Eq. (5.35)) as

$$\begin{aligned}
K[W(\mathbf{x},\theta+d\theta)||P(\mathbf{x},\theta)] &= \int W(\mathbf{x},\theta+d\theta)\ln\left(\frac{W(\mathbf{x},\theta+d\theta)}{W(\mathbf{x},\theta)}\right)d^2\mathbf{x} \\
&= \int [W(\mathbf{x},\theta)+\partial_\theta W(\mathbf{x},\theta)d\theta+\frac{1}{2}\partial_\theta^2 W(\mathbf{x},\theta)d\theta^2+\mathcal{O}(d\theta^3)]\times \\
&\times\left[\frac{\partial_\theta W(\mathbf{x},\theta)}{W(\mathbf{x},\theta)}d\theta+\frac{1}{2}\frac{\partial_\theta^2 W(\mathbf{x},\theta)}{W(\mathbf{x},\theta)}d\theta^2+\mathcal{O}(d\theta^3)\right]d^2\mathbf{x} \\
&= \int\frac{(\partial_\theta W(\mathbf{x},\theta)d\theta)^2}{W(\mathbf{x},\theta)}d^2\mathbf{x}+\mathcal{O}(d\theta^3) \\
&= \frac{1}{2}\mathcal{I}_W(\theta)d\theta^2+\mathcal{O}(d\theta^3),
\end{aligned} \tag{5.41}$$

where we identify $\mathcal{I}_W(\theta)$ as the Wigner-Fisher information, that can be written in terms of the mean vector and the covariance matrix resulting in

$$\mathcal{I}_W(t) = \dot{\mathbf{v}}_t^\dagger\Theta_t^{-1}\dot{\mathbf{v}}_t+\mathrm{Tr}[(\Theta_t^{-1}\dot{\Theta}_t)^2]. \tag{5.42}$$

Note that we changed $\theta = t$ because in thermal kinematics the interesting parameter is time, and, as a result, the thermal kinematics variables become

$$v_W(t) = \sqrt{\frac{1}{2}\mathcal{I}_W(\rho(t))} \tag{5.43}$$

$$\mathcal{L}_W(t) = \int_0^t\sqrt{\frac{1}{2}\mathcal{I}_W(\rho(t))} \tag{5.44}$$

$$\varphi_W(t_s) = \frac{\mathcal{L}_W(t_s)}{\mathcal{L}_W(t_f)}. \tag{5.45}$$

Additionally, the Wigner-Fisher information can be separated into a passive and ergotropic contribution: $\mathcal{I}_W(t) = \mathcal{I}_\pi(t)+\mathcal{I}_\mathcal{E}(t)$. To see this, we will use that the passive state of any Gaussian state is always thermal, so the first term of Eq. (5.42) is zero because $\mathbf{v}_\pi = 0$. Then, using that the covariance matrix is $\Theta_\pi = f(\beta_\pi)\mathbb{I}$ where $f(\beta_\pi) = \sqrt{|\Theta|}$, we find $\Theta_\pi^{-1} = (1/f(\beta_\pi))\mathbb{I}$ and $\dot{\Theta}_\pi = f(\beta_\pi)\dot{S}_W(t)\mathbb{I}$. Substituting these matrices in the second term of Eq. (5.42), we obtain the passive contribution in the Wigner-Fisher information is

$$\begin{aligned}
\mathcal{I}_\pi(t) &= \mathrm{Tr}[(\Theta_t^{-1}\dot{\Theta}_t)^2] \\
&= \mathrm{Tr}\left[\left(\frac{1}{f(\beta_\pi)}f(\beta_\pi)\dot{S}_W(t)\mathbb{I}\right)^2\right] \\
&= 2\dot{S}_W^2(t)
\end{aligned} \tag{5.46}$$

With this, we introduced all the necessary quantities and can start discussing our results. First, we will discuss the heating-cooling asymmetry for thermal states in Sec. 5.3. Then we will investigate the consequences of displacing this thermal state in Sec. 5.4 and squeezing this thermal state in Sec. 5.5. Because a squeezed state relaxate faster than a displaced state, we will try to invert the asymmetry by combining these Gaussian operations in Sec. 5.6.

5.3 Asymmetry for Thermal States

In this section, we consider a Gaussian state initially in a thermal state with inverse temperature β_i interacting with a bath at inverse temperature β_f. After solving Eq. (5.19) and Eq. (5.26), we find the dynamics is described by a zero mean vector ($\mathbf{v}_{th} = 0$) and a covariance matrix equal to $\Theta_{th} = \Delta_\beta \mathbb{I}$, being $\Delta_\beta = [f(\beta_i) - f(\beta_f)]e^{-\Gamma t} + f(\beta_f)$ where Γ is the dissipation rate, and $f(\beta) = \bar{n} + 1/2$.

Similar to the qubits case, to find two thermodynamically equivalent initial states, we use the Wigner relative entropy defined in Eq. (5.35). Using that for thermal states $|\Theta| = (\Delta_\beta)^2$ and $\Theta_{th}^{-1} = \frac{1}{\Delta_\beta}\mathbb{I}$, being the equilibrium $\lim_{t\to\infty} \Delta_\beta = f(\beta_f)$, the analytical expression for the Wigner-Fisher information (defined in Eq. (5.42)) is

$$\mathcal{I}_{W_{th}}(t) = 2\left[\frac{\partial}{\partial t}\ln \Delta_\beta\right]^2, \tag{5.47}$$

the statistical velocity (defined in Eq. (5.43))

$$v_{W_{th}}(t) = \frac{\partial}{\partial t}\ln \Delta_\beta, \tag{5.48}$$

and the statistical length (defined in Eq. (5.44))

$$\mathcal{L}_{W_{th}}(t) = \ln\left(\frac{[f(\beta_i) - f(\beta_f)]e^{-\Gamma t} + f(\beta_f)}{f(\beta_i)}\right). \tag{5.49}$$

The statistical length in the equilibrium is

$$\mathcal{L}^{eq}_{W_{th}} \equiv \lim_{t\to\infty} \mathcal{L}_{W_{th}}(t) = \ln\left[\frac{f(\beta_f)}{f(\beta_i)}\right]. \tag{5.50}$$

Because of the explicit dependence on β_i, two states with different initial temperatures that relaxate to the same final temperature will have different statistical lengths. To simplify the notation, we define $f_i \equiv f(\beta_f)/f(\beta_i)$ and $k = \exp(-\Gamma t)$ and use the expressions (5.49) and (5.50) to write the degree of completion (defined in Eq. (5.45)) as

$$\varphi_{W_{th}}(t) = \frac{\ln(k + (1-k)f_i)}{\ln(f_i)} \equiv \varphi_i(k). \tag{5.51}$$

Now, we can analyse the thermal relaxation asymmetry for thermal states. We know that $\beta_c > \beta_f > \beta_h$, which results in $f(\beta_c) < f(\beta_f) < f(\beta_h)$, leading to

$$1 > \frac{f(\beta_f)}{f(\beta_h)} = f_h \tag{5.52}$$

$$1 < \frac{f(\beta_f)}{f(\beta_c)} = f_c, \tag{5.53}$$

which is equivalent to $\ln(f_c) > 0$ and $\ln(f_h) < 0$. Next, we will use the Jensen inequality for $f(x) = \ln(x)$, that is $f(kx_1 + (1-k)x_2) \geq kf(x_1) + (1-k)f(x_2)$. In our case, we can use $x_1 = 1$ and $x_2 = f_i$ to obtain

$$\ln(k + (1-k)f_i) \geq (1-k)\ln(f_i), \tag{5.54}$$

where we have the numerator of the degree of completion on the left side and the denominator on the right side. For the heating case ($i = c$), $\ln(f_c) > 0$ so dividing Eq. (5.54) by it do not change the inequality and we find that the degree of completion (Eq. (5.51)) for this process is

$$\varphi_c = \frac{\ln(k + (1-k)f_c)}{\ln(f_c)} \geq 1 - k. \tag{5.55}$$

On the other hand, for the cooling case ($i = h$), $\ln(f_h) < 0$ so the division invert the inequality in Eq. (5.54) and we find that the degree of completion for this process is

$$\varphi_h = \frac{\ln(k + (1-k)f_h)}{\ln(f_h)} \leq 1 - k. \tag{5.56}$$

By combining both degrees of completion inequalities, we obtain that $\varphi_c \geq \varphi_h$, which means that heating is always faster or equal to cooling a Gaussian thermal state.

It is relevant to study whether the presence of thermal relaxation asymmetry in Gaussian systems is also explained by the difference in entropy production rate. The Wigner relative entropy for a thermal state is

$$K[W_{th}||W_f] = -1 + \frac{\Delta_\beta}{f(\beta_f)} + \ln\frac{f(\beta_f)}{\Delta_\beta}. \tag{5.57}$$

By taking the negative time derivative of Wigner relative entropy, we find that the entropy production rate is

$$\Pi_{W_{th}} = \frac{\dot{\Delta}_\beta}{\Delta_\beta} - \frac{\dot{\Delta}_\beta}{f(\beta_f)} = \dot{S}_{W_{th}}\left(1 - \frac{\Delta_\beta}{f(\beta)}\right), \tag{5.58}$$

where $\dot{S}_{W_{th}} = \dot{\Delta}_\beta/\Delta_\beta$ is the time derivative of the Wigner entropy (Eq. (5.33)). From this expression, we obtain Fig. 11 by using $K[W_c||W_f] = K[W_h||W_f] = 0.5$ and $T_f = 2$ (in natural units). Note that heating entropy production reaches zero before cooling, corroborating our conclusion that heating relaxes faster than cooling. Also, heating has an initially higher entropy production rate than cooling, which explains the asymmetry.

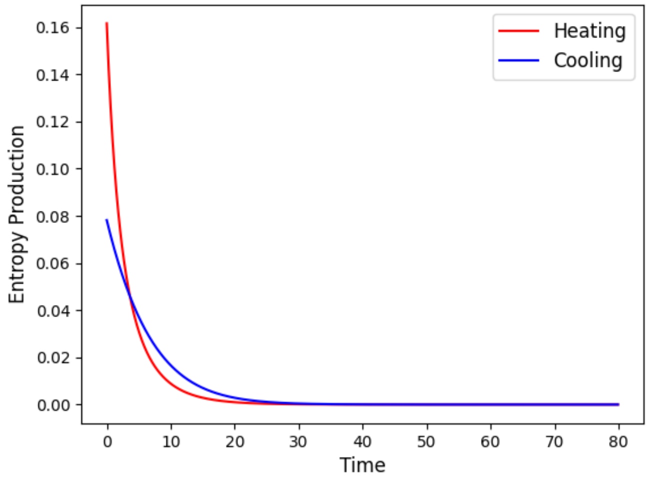

Figure 11 – Entropy production rate for heating (red) and cooling (blue) thermal states.

5.4 Asymmetry for Displaced States

Now, we can analyse what happens to the asymmetry when we displace a thermal state. The solution of Eq. (5.19) and Eq. (5.26) is that the mean vector becomes $\vec{v}_d = (\mu_t, \mu_t^*)$ where $\mu_t = \mu e^{-(i\omega+\Gamma/2)t}$ while the covariance matrix remains unchanged, i.e., $\Theta_d = \Delta_\beta \mathbb{I}$. Using these results in the Wigner relative entropy (Eq. (5.35)), we find

$$K[W_d||W_f] = -1 + \frac{|\mu|^2}{f(\beta_f)}e^{-\Gamma t} + \frac{\Delta_\beta}{f(\beta_f)} + \ln\frac{f(\beta_f)}{\Delta_\beta}. \tag{5.59}$$

The term containing the displacement parameter (μ) is independent of the system's initial temperature, which means that, regardless of the initial temperature, applying a displacement operator to any thermal state results in the same shift in their Wigner relative entropy. Thus, for simplicity, we chose two equidistant thermal states and then applied the same amount of work on both (which is equivalent to using the same μ). However, because displacement also adds ergotropy to the state, it is important to use initial states with the same ergotropy. The ergotropy for a displaced state is $\mathcal{E}_d = |\mu|^2 e^{-\Gamma t}$, which means that using the same μ is enough to guarantee different states have the same ergotropy. As we did before, we could try to analytically calculate the thermal kinematics variables, but the Wigner-Fisher information for a displaced state is

$$\mathcal{I}_{W_d}(t) = \frac{2}{\Delta_\beta}\left[\frac{\dot{\Delta}_\beta^2}{\Delta_\beta} + \left(\omega^2 + \frac{\Gamma^2}{4}\right)|\mu|^2 e^{-\Gamma t}\right], \tag{5.60}$$

and, despite being possible to analytically find the statistical velocity and length, the complexity of the final expressions ultimately makes it difficult to extract useful information. Therefore, we use a numerical approach for this case. The chosen parameter were $\omega = 1$,

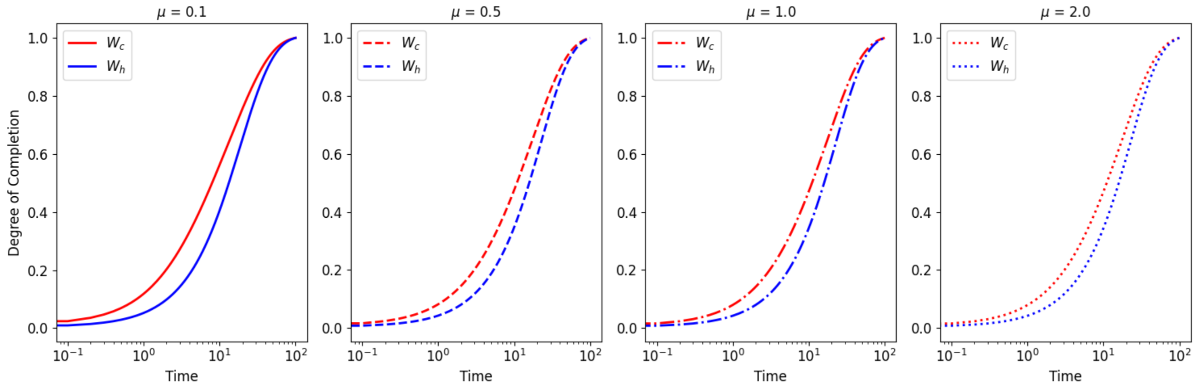

Figure 12 – Degree of completion for a displaced thermal state. Each graph corresponds to a different displacement parameter (μ).

$\Gamma = 0.1$, $T_f = 2$, and $K[W_c||W_f] = K[W_h||W_f] = 0.5$. Fig. 12 shows the degree of completion for different displacement parameters, with the thermal relaxation of W_c (in red) being faster than that of W_h (in blue) for all the displacements. An interesting result is that increasing the displacement parameter, moves the initial state further away from the equilibrium (as one can see from Eq. (5.59)), but the heating-cooling asymmetry remains unchanged. However, the initial statistical velocity also increase with μ as shown in Fig. 13 and expected from Eq. (5.60). Therefore, a possible explanation for the asymmetry remaining unchanged with displacement is that the initial distance from equilibrium is compensated by a higher initial velocity.

Another way to analyse this asymmetry is using the entropy production. Given that displacement generates ergotropy in the system, we can use the entropy production separation into passive and ergotropic contribution from Eq. (5.40). The total entropy

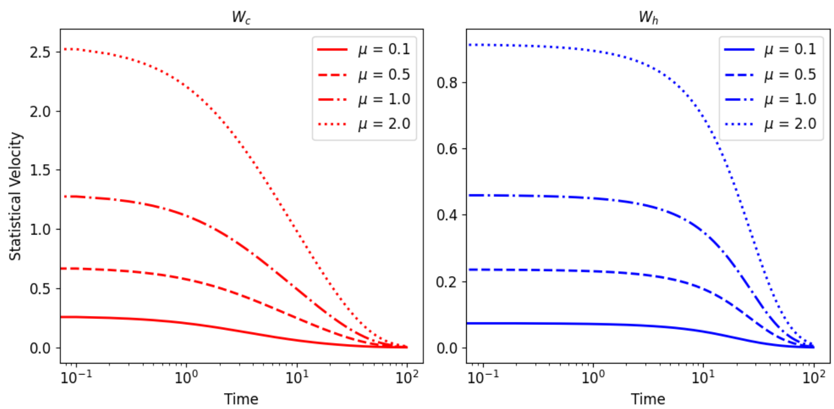

Figure 13 – Statistical velocity of a displaced thermal state as a function of time. Each curve corresponds to a different displacement parameter (μ).

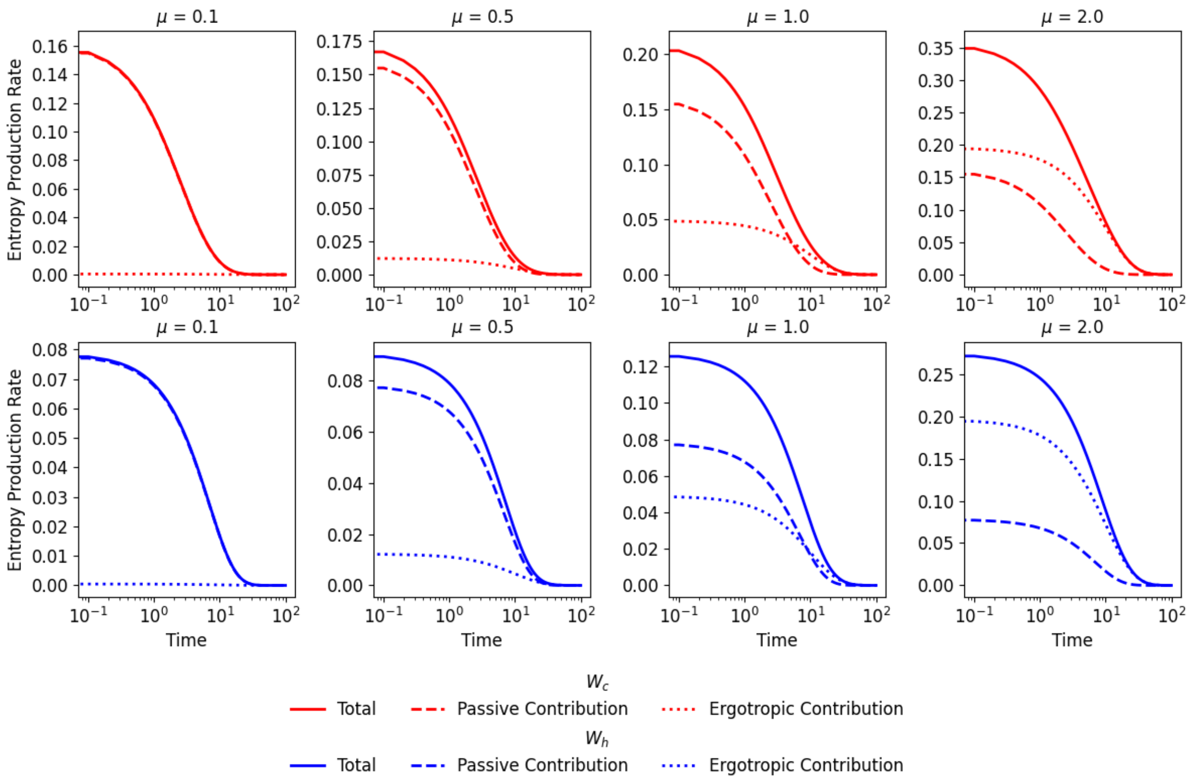

Figure 14 – Total entropy production rate(solid) with its ergotropic (dotted) and passive (dashed) contributions for heating (red) and cooling (blue) equidistant displaced thermal states.

production rate for a displaced state is given by

$$\Pi_{W_d} = \frac{1}{f(\beta)}(\Gamma|\mu|^2 e^{-\Gamma t} - \dot{\Delta}_\beta) + \frac{\dot{\Delta}_\beta}{\Delta_\beta}. \tag{5.61}$$

The entropy production of the passive state is

$$\Pi_{W_\pi} = \dot{S}_{W_\pi}\left(1 - \frac{f(\beta_\pi)}{f(\beta)}\right) = \Pi_{W_{th}}, \tag{5.62}$$

which is the same entropy production of the thermal state (Eq. (5.58)), because displacement do not change the covariance matrix, resulting in $f(\beta_\pi) = \Delta_\beta$ and $\dot{S}_{W_\pi} = \dot{S}_{W_{th}}$. Finally, the ergotropic contribution is in terms of the ergotropic loss rate: $\dot{\mathcal{E}}_d = -\Gamma|\mu|^2 e^{-\Gamma t}$. Fig. 14 shows the total entropy production rate (solid line), its passive contribution (dashed line) and ergotropic contribution (dotted line). Each plot has an increasing displacement parameter μ from left to right, being W_c in red and W_h in blue. In all the graphs, W_c starts with a higher total entropy production rate than W_h, which again explains why the thermal relaxation of W_c is faster. In addition, making greater displacements increases the ergotropic contribution while keeping the passive contribution unchanged. As a consequence, the total entropy production increases with μ because of the ergotropic contribution. Another interesting result is that, when comparing W_c and W_h with same μ, the ergotropic contribution do not change, meaning the passive contribution is the generator of asymmetry that is unaffected by the displacement operation, which agrees with our previous discussion.

5.5 Asymmetry for Squeezed States

In this section, we study the thermal relaxation asymmetry of squeezed thermal states. Unlike the displacement operation, squeezing a thermal state preserves the zero mean vector and changes the covariance matrix to

$$\Theta_s = f(\beta_t)\begin{bmatrix} \cosh(2r_t) & -e^{i\phi_t}\sinh(2r_t) \\ -e^{-i\phi_t}\sinh(2r_t) & \cosh(2r_t) \end{bmatrix}, \tag{5.63}$$

where ϕ_t is the squeezing phase, the squeezing parameter is implicitly defined as

$$\cosh(2r_t) = \frac{1}{f(\beta_t)}[\Delta_\beta + 2f(\beta_i)e^{-\Gamma t}\sinh^2(r)], \tag{5.64}$$

and β_t is implicitly defined by

$$f(\beta_t) = \sqrt{\Delta_\beta^2 + 4f(\beta_i)f(\beta)e^{-2\Gamma t}(e^{\Gamma t}-1)\sinh^2(r)}. \tag{5.65}$$

With the information above, the relative entropy for a squeezed state is

$$K[W_s||W_f] = -1 + \frac{f(\beta_t)}{f(\beta)}\cosh(2r_t) + \ln\left(\frac{f(\beta)}{f(\beta_t)}\right). \tag{5.66}$$

Different from the displacement case, because the r_t and $f(\beta_t)$ that contains the initial temperature are related, doing the same squeezing in different thermal states do not preserve their equidistance. As a consequence, to make a fair comparison, we must use different squeezing parameters for each state. Just like displacement, squeezing also generates ergotropy in the state, so it is important to choose initial states with same ergotropy. The ergotropy of a squeezed thermal state is

$$\mathcal{E}_s(t) = \omega f(\beta_t)(\cosh(2r_t) - 1). \tag{5.67}$$

Because of the dependence of $f(\beta_t)$, in $t = 0$, $f(\beta_t) = f(\beta_i)$ and applying the same squeezing (that is, the same r and ϕ) in different thermal states results in a higher ergotropy to the squeezed hotter state. For clarity, we will describe how we choose a state with same initial relative entropy and ergotropy. To start, we choose an initial relative entropy K_0 and a squeezing parameter for the hotter state (r_h), for example. Then, we look for the temperature T_h a thermal state must have to satisfy $K[W_h||W_f] = K_0$ after squeezing with parameter r_h. After finding our hotter initial state, we can try to find the colder initial state. By

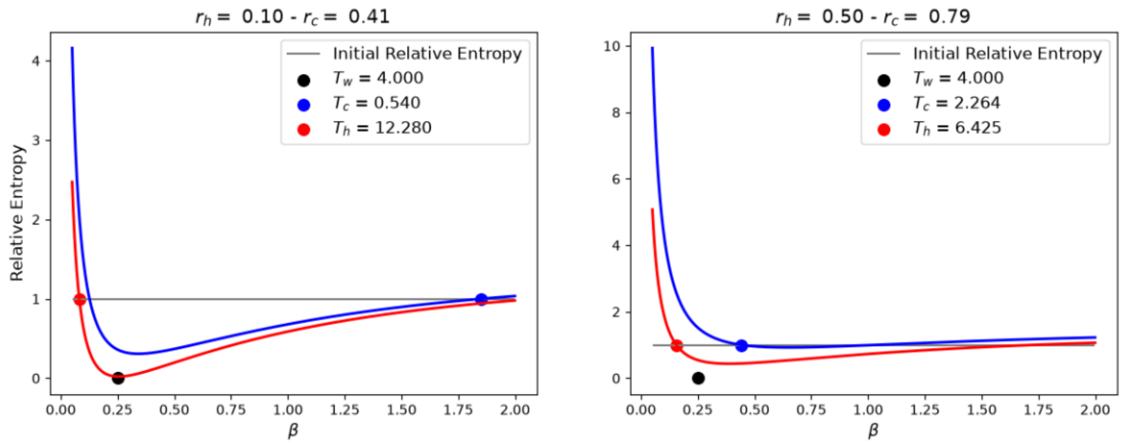

Figure 15 – Wigner relative entropy of a squeezed thermal state and the equilibrium thermal state as a function of inverse temperature. The black line indicates the chosen initial Wigner relative entropy that both hot (red dot) and cold (blue dot) squeezed states must have. The left plot uses a small squeezing parameters and the right plot use bigger squeezing parameters.

combining Eq. (5.66) and Eq. (5.67), we get that $f(\beta_c)-f(\beta_h)+f(\beta_f)\ln(f(\beta_h)/f(\beta_c))=0$ must be true to satisfy the condition of same initial relative entropy and ergotropy. Given that we already know T_h, we can solve this equation to obtain the temperature of the cold squeezed state and use Eq. (5.64) to find r_c. We will call the initial hotter state W_h and the initial colder state W_c. Squeezed states do not have a real temperature, but because the passive state is always a thermal state, we will assume that W_h will cool down and W_c will heat up. Fig. 15 presents Wigner relative entropy as a function of temperature and shows which initial temperatures we find in the end of this process. The left plot corresponds to the smaller squeezing parameters and the right plot corresponds to the bigger squeezing parameters. Note that increasing the squeezing parameter almost conserves de difference between r_c and r_h. However, the colder temperature (in blue) have a bigger variation than the hotter temperature (in red), resulting in a bigger variation of ergotropy. From this, it seems that colder states are more sensible to variations on the squeezing parameter.

The Wigner-Fisher information of a squeezed thermal state is

$$\mathcal{I}_{W_s}(t)=2\dot{S}_{W_s}^2+8\dot{r}_t^2+2\dot{\phi}_t^2\sinh^2(2r_t), \tag{5.68}$$

where the time derivative of the Wigner entropy is

$$\dot{S}_{W_s}=\frac{\dot{f}(\beta_t)}{f(\beta_t)}. \tag{5.69}$$

Again, obtaining the analytical thermal kinematic expressions from Eq. (5.68) will not bring additional information, so we will solve it numerically. In Fig. 16, we show the

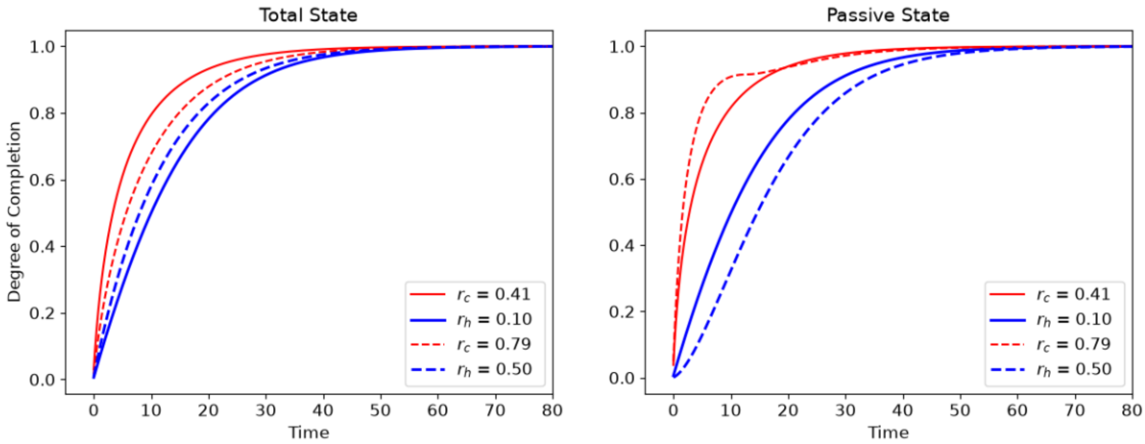

Figure 16 – Degree of completion for squeezed thermal states (left) and their passive states (right). We compared a smaller squeezing parameter (solid lines) with a bigger squeezing parameter (dashed lines) for both initially colder (r_c) and hotter (r_h) states.

degree of completion as a function of time for the squeezed thermal state (left) and its passive state (right). In each plot, we compared a squeezing with small parameters (solid lines) with a squeezing with bigger parameters (dashed lines). In the both plots, because W_c (in red) is always above W_h (in blue), we can say that heating is faster than cooling for these squeezed states and their passive states. However, in the left plot, by comparing the solid lines with the dashed lines, we observe that increasing the squeezing parameters reduces the heating-cooling asymmetry of the total state. However, in the right plot, the asymmetry in the passive states seems to increase as we increase the squeezing parameters.

To investigate this behaviour, we will analyse the entropy production rate and separate into its passive and ergotropic contribution from Eq. (5.40). The total entropy production rate is

$$\Pi_{W_s} = \dot{S}_{W_s} - \frac{\dot{E}}{\omega f(\beta_f)}, \tag{5.70}$$

where the time derivative of the Wigner entropy was calculated in Eq. (5.69) and the last term is the time derivative of the energy,

$$\dot{E} = \omega[\dot{f}(\beta_t)\cosh(2r_t) + 2\dot{r}_t f(\beta_t)\sinh(2r_t)]. \tag{5.71}$$

The passive state have $f(\beta_\pi) = f(\beta_t)$, resulting in the entropy production

$$\Pi_{W_\pi} = \dot{S}_{W_s}\left(1 - \frac{f(\beta_t)}{f(\beta_f)}\right), \tag{5.72}$$

which depends on the squeezing parameter, so it will change for all of our curves. According to Eq. (5.40), to find the ergotropic contribution, we take the time derivative of Eq. (5.67) and find

$$\frac{\dot{\mathcal{E}}_s}{\omega f(\beta_f)} = \frac{\dot{f}(\beta_t)}{f(\beta_f)}(\cosh(2r_t) - 1) + 2\frac{f(\beta_t)}{f(\beta_t)}\sinh(2r_t)\dot{r}_t. \tag{5.73}$$

As this equation depends on the squeezing parameter and initial temperature, we find out the ergotropic contribution will also change for all of our curves.

Fig. 17 show the entropy production rate, where the upper plots correspond to smaller r and the lower plots correspond to bigger r. Each graph shows the total entropy production rate (solid line) with its passive (dashed line) and ergotropic (dotted line) contributions. First, the initial total entropy production of W_c is bigger than the initial total entropy production of W_h, which explain why heating is faster than cooling again. Also, when we increase the squeezing parameters, the heating and the cooling total entropy productions decrease, but the heating process suffers a bigger reduction when compared with cooling. This can be explained by the previous discussion that colder states have

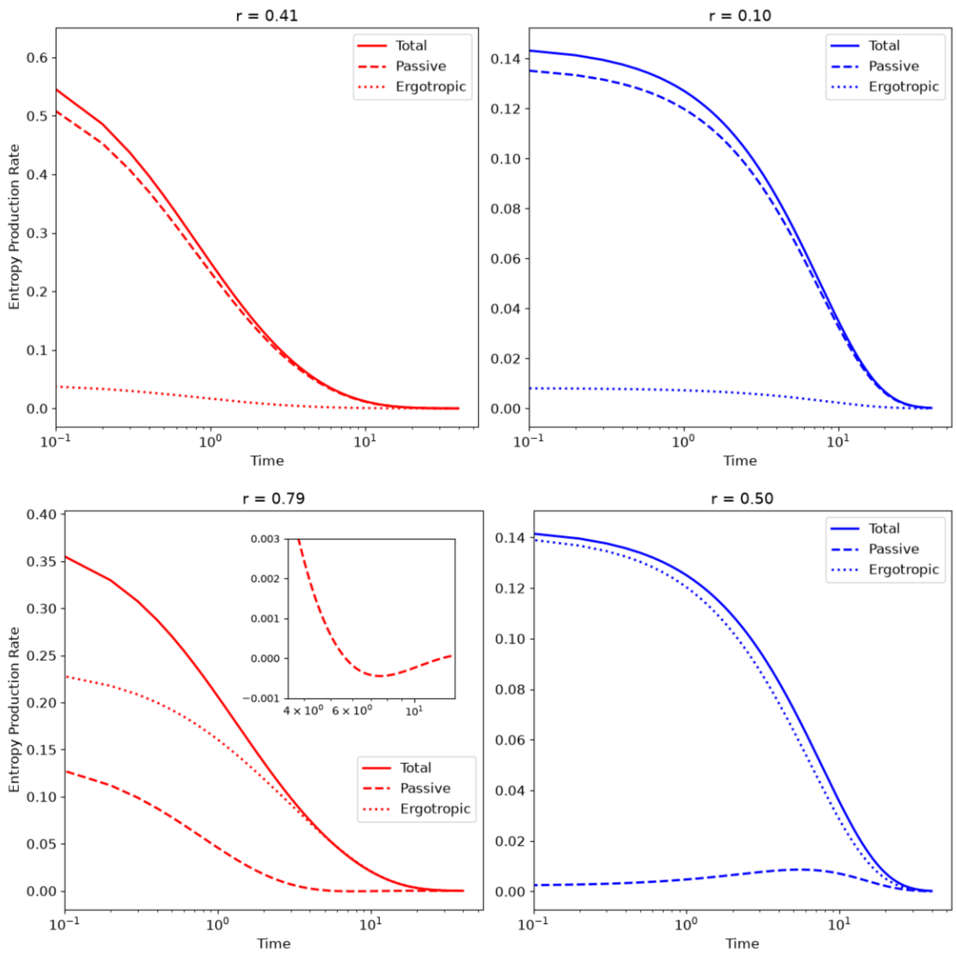

Figure 17 – Entropy production rate for heating (red) and cooling (blue) squeezed thermal states with smaller parameters (upper plots) and bigger parameters (lower plots). We separate the passive contribution (dashed) and ergotropic contribution (dotted) of the total entropy production rate (solid). The inset plot show the region where the passive state entropy production becomes negative.

higher sensibility to squeezing variations. As a result, for bigger squeezing parameters, the difference in initial total entropy production between heating and cooling is smaller, which explains why the asymmetry was reduced for the total squeezed state. However, as we already noticed in the equations of entropy production, unlike the displacement operation, squeezing affects both passive and ergotropic contributions, which means that the asymmetry is not fully generated by the passive state. When we look at the passive contribution, the difference between the initial entropy production of heating and cooling becomes bigger as we increase the squeezing parameter. This can explain why the asymmetry in the passive contribution increases. The explanation of how these related quantities can have opposite behaviours with the increase of squeezing parameter is in the ergotropic contribution. Given that for small r the passive contribution was bigger, to preserve the same entropy production, the ergotropic contribution should compensate the decrease of the passive. However, this is not the case and ergotropic increase do not compensate the passive loss as shown in Fig. 17.

One last comment of this section is regarding the oscillation in the degree of completion that appears for the passive state of the big squeezing parameter case (red dashed line of the right plot in Fig. 16). Additionally, the statistical velocity of this same passive state is shown in Fig. 18 and also presents this oscillation. Recalling the degree of completion and statistical velocity we had for the interacting qubits, we also saw this oscillations and attributed them to a local non-Markovian dynamics witnessed as negative entropy production. The inset in Fig. 17 shows that the entropy production of the passive state with big squeezing parameter also have negative value for a period of time, which is a signature of non-Markovianity. From this observations, we can again affirm that non-Markovianity may be related to faster relaxation. The origin of this non-Markovian dynamics for the passive state can be understood as a part of squeezing being converted into energy for the passive state. It is interesting that this behaviour was only observed for heating state with big squeezing parameter and not in the correspondent cooling state. One possibility is that the squeezing parameter used in heating was bigger than the one in cooling. However, this cooling process do have a different behaviour, which is a small increase in its entropy production and statistical velocity. This means that, instead of receiving a feedback information from the environment, the system accelerates and loose more information to the environment. We do not know its cause, but it is important to highlight that, while negative entropy production indicate non-Markovianity, the opposite is not always true, i.e., non-Markovian dynamics without negative entropy production is possible (54).

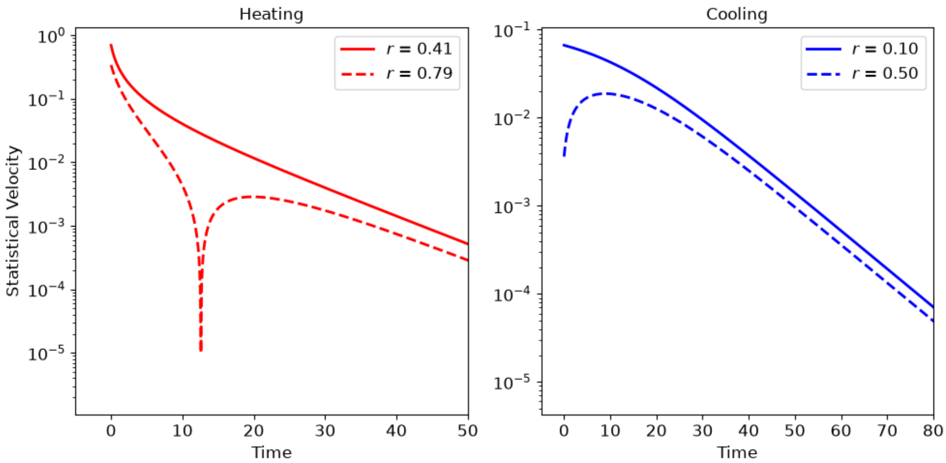

Figure 18 – Statistical velocity of the heating (red) and cooling (blue) passive states of squeezed states with smaller parameter (solid lines) and bigger parameters (dashed lines).

5.6 Asymmetry Inversion for Gaussian States

Until this moment, we only observed that heating was faster than cooling, but previous works have shown that this is not always true (7, 14, 15). In this section, we will try to invert this asymmetry using the previous observation that squeezed states relaxate faster than displaced states. So, we will apply a squeezing operation on the slower process, that is cooling, and will apply a displacement operation on heating because both initial state must have the same initial ergotropy.

To find a hotter squeezed state and colder displaced state with the same initial relative entropy and ergotropy, we will use a similar approach as the one from Sec. 5.5. We combine $K[W_s||W_f] = K[W_d||W_f]$ and $\mathcal{E}_s = \mathcal{E}_d$ to obtain $f(\beta_d) - f(\beta_s) + f(\beta_f)\ln(f(\beta_s)/f(\beta_d)) = 0$, which can be used to find the temperature of a state from the the other state.

The degree of completion for this system is presented in Fig. 19, and shows that it is possible to invert the asymmetry depending on the initial state condition. In the left graph, the displacement and squeezing parameters are smaller than in the right graph, and the asymmetry inversion is partial because cooling is faster only after some time. In the right graph, where the parameters are bigger, cooling is faster than heating for all times. Fig. 20 show the degree of completion of the passive states. Despite the inversion of the total state, the passive states still have heating faster than cooling as expected from the thermal states results in Sec. 5.3. Also, we notice that increasing the ergotropy of the total state, makes the passive relaxation slower, but it seems to preserve the asymmetry between heating and cooling.

To study the entropy production rate of heating the displaced state, we used

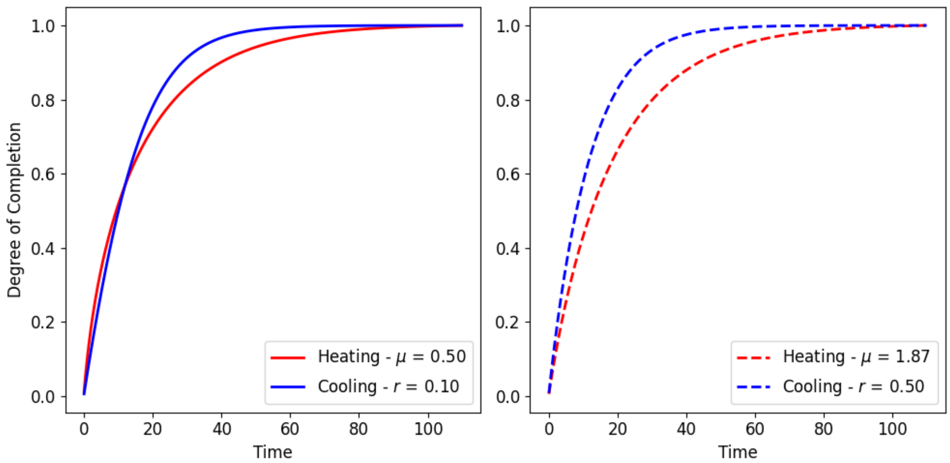

Figure 19 – Degree of completion for heating the displaced state (red) and cooling the squeezed state (blue).

the same expressions of Section 5.4 and, for the entropy production rate of cooling the squeezed state, we used the expressions of Section 5.5. Fig. 21 shows the total entropy production (solid lines), the passive (dashed lines) and the ergotropic (dotted lines) contributions for the entropy production of smaller and bigger parameters, respectively. Given that heating is faster for the passive states, what accelerates cooling to create the inversion must be the ergotropic contribution. Different from the results comparing only squeezing or only displacement, the ergotropic contribution of cooling (squeezing) is initially higher than the one of heating, which explains the full inversion of the asymmetry. Furthermore, for the partial inversion, we see that the ergotropic part had almost no participation in the relaxation and all the asymmetry was generated by the passive state. In this case, heating had higher initial entropy production and was faster than cooling according to the degree of completion in Fig. 19. However, when the entropy production of cooling

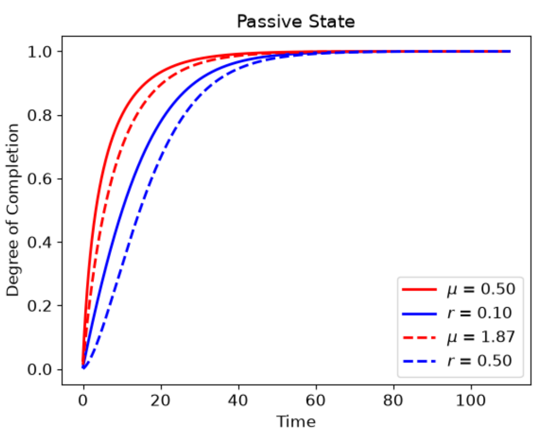

Figure 20 – Degree of completion of the passive state of the heating displaced state (red) and of the cooling squeezed state (blue).

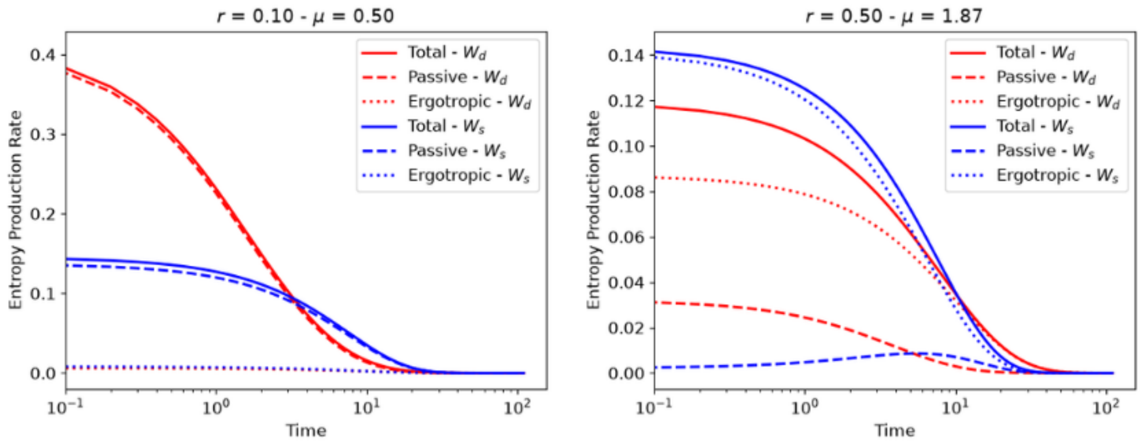

Figure 21 – Left plot shows the entropy production rate of smaller μ and r. Right plot shows the entropy production rate of bigger μ and r. Heating the displaced state is in red and cooling the squeezed state is in blue. The passive contribution is in dashed lines, the ergotropic contribution is in dotted lines, and the total entropy production rate is in solid lines.

cross with heating and we find the partial inversion in the degree of completion. In this case, the initial entropy production could not tell us which process relaxate faster, but its time evolution is still helpful to understand the dynamics. Additionally, heating crossing cooling at some time was already observed in the ergotropic Mpemba effect (9) and it is a consequence of comparing the relaxation of a displaced state with a squeezed state.

6 CONCLUSION

The main question investigated in this work was how quantum properties affect the heating-cooling asymmetry in discrete and continuous-variable systems. The relaxation dynamics was described by the Lindblad master equations and we used the thermal kinematics approach to identify the heating-cooling asymmetry. In this respect, we introduced an adaptation of thermal kinematics to Gaussian systems. In addition to thermal kinematics, we analysed how entropy production can be used to explain which thermal relaxation process is faster.

For the discrete case, the two-interacting-qubits model allowed us to study the thermal relaxation of systems with coherence and entanglement. By applying Davies maps to this model, the global state always thermalizes to the bath's temperature, but the interaction between the qubits can generate correlations between them, and, as a result, we cannot write the global steady state as the tensor product of qubit's thermal states. However, we can associate an effective temperature to the qubits with the cost of this temperature at equilibrium being lower than the bath's temperature. Another consequence of the qubits interaction generating correlations is that the local entropy production rate becomes negative at some periods, which indicates the a local non-Markovian dynamics. We noticed that the thermal relaxation case with non-Markovianity was faster than the no-interaction and full Markovian case. However, we observed that strong interaction causes entanglement, which reduces the asymmetry and make the relaxation slower than the weaker interaction without entanglement generation. Additionally, we saw that initial global coherence can apparently increase the asymmetry, changing the intermediary dynamics, but it do not affect the total equilibration time of the qubits.

For the continuous-variable case, we explored the heating-cooling asymmetry in single-mode Gaussian states. We considered that squeezing can generate quantum resources, such as coherences, in Gaussian states. However, instead of directly quantifying these resources, we used ergotropy to separate how much of the nonequilibrium energy is thermodynamically useful. Because the passive state of Gaussian systems is always a thermal state, any quantumness of the system will be related to ergotropic quantities. An interesting consequence of using the ergotropic approach is the possibility of separating the Wigner-Fisher information and the entropy production rate into passive and ergotropic contributions. So, we can separate the asymmetry origin from passive contributions (that have no quantumness for Gaussian systems) and from ergotropic contribution (that may have quantumness).

The first Gaussian system we analysed was thermal states, in which we analytically showed that heating is always faster than cooling and where entropy production successfully explained its origin. Second, we compared the relaxation of different thermal states that were equally displaced and showed that displacement does not affect the asymmetry, being the passive contribution its generator. Again, the total entropy production explained heating being faster than cooling and separating the contributions showed that the ergotropy was the same for both heating and cooling, which is a consequence of being independent of the system's initial temperature. Then, we study the asymmetry in squeezed thermal states where both ergotropic and passive contributions are affected by the Gaussian operation. In this case, using bigger squeezing parameters resulted in a reduction of the asymmetry of the total state while the passive state had its asymmetry increased. To explain this apparent contradiction, we observed that higher squeezing parameter reduced the initial entropy production of the hot and cold passive states, but it increase their difference. However, we noticed that increasing the initial ergotropy did not lead to enough increase in the ergotropic contribution of entropy production to compensate the passive reduction. As a result, the difference of the heating and cooling total entropy production decreases, causing the asymmetry reduction. A last result from the asymmetry with squeezed states it that we identified a non-Markovian dynamics for the heating passive state of the big squeezing state and we also observed an acceleration of the dynamics. As a result, once again we have an indication that non-Markovian dynamics can help a faster relaxation. Until this moment, all results showed that heating was faster than cooling, but it is known from literature that this asymmetry can be inverted. This motivated the investigation of how we could have cooling faster than heating in our systems. Using the knowledge that squeezed states relaxate faster than displaced states, we squeezed the hotter state and displaced the colder state. With this combination we inverted the asymmetry partially (for smaller parameters) and totally (for bigger parameters). We learned that the full inversion is generated only by the ergotropic contribution, while heating is still faster than cooling for the passive states. Also, we discussed that the partial inversion is actually a consequence of squeezing having a faster relaxation than displaced states, which was already discussed in the ergotropic Mpemba effect (9). In this case, the initial entropy production failed to indicate the faster process, but its time evolution helped understand the partial inversion.

Beyond the investigations done in this work, there is a variety of quantum properties and systems where heating-cooling asymmetry could be generated. Regarding the same model of two-interacting qubits, there is also the possibility of investigating other interaction hamiltonians, such as XXX, XXZ, XYZ, and transverse field Ising. In the context of Gaussian states, the ergotropic approach was not a clear indicator of quantumness. So, a next step can be quantifying the quantum resources, such as coherence, to understand how much of the behaviour we studied is actually a consequence of quantum properties. In this respect, extending the analysis for more modes can allow the investigation of the heating-cooling asymmetry in entangled Gaussian systems. As we briefly discussed in Sec. 5.5, the passive state of squeezed states can have a non-Markovian dynamics that seems to accelerate the relaxation. So, further investigating this aspect can also be an interesting direction. Another possible future direction would be to study why entropy production is asymmetric. A possible direction is using the Onsager relations for far-from-equilibrium Gaussian states as derived in (59). The Onsager relations write the entropy production in terms of currents and Ref. (59) showed that it becomes non-linear and a not even function for arbitrarily far-from-equilibrium Gaussian states.

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APPENDIX A – CONCURRENCE OF AN X-STATE

A general way to define an X-state is

$$\rho = \begin{bmatrix} a & 0 & 0 & f \\ 0 & b & e & 0 \\ 0 & e^* & c & 0 \\ f^* & 0 & 0 & d \end{bmatrix}. \tag{A.1}$$

As we presented in Sec. 4.1, the concurrence of two qubits is defined as

$$C(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \tag{A.2}$$

where $\lambda_i > \lambda_j$ for $i < j$ are the square root of the eigenvalues of the matrix $R = \rho\tilde{\rho}$ with $\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$ and ρ^* being the complex conjugate of the density matrix. To start computing the concurrence of an X-state, we first find that

$$\rho^* = \begin{bmatrix} a & 0 & 0 & f^* \\ 0 & b & e^* & 0 \\ 0 & e & c & 0 \\ f & 0 & 0 & d \end{bmatrix}. \tag{A.3}$$

Using that

$$\sigma_y \otimes \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \tag{A.4}$$

it is possible to compute

$$\begin{aligned} \tilde{\rho} &= (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y) \\ &= \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & 0 & 0 & f^* \\ 0 & b & e^* & 0 \\ 0 & e & c & 0 \\ f & 0 & 0 & d \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} d & 0 & 0 & f \\ 0 & c & e & 0 \\ 0 & e^* & b & 0 \\ f^* & 0 & 0 & a \end{bmatrix}. \end{aligned} \tag{A.5}$$

As a result, we find

$$
\begin{aligned}
R &= \rho\tilde{\rho} \\
&= \begin{bmatrix} a & 0 & 0 & f \\ 0 & b & e & 0 \\ 0 & e^* & c & 0 \\ f^* & 0 & 0 & d \end{bmatrix} \begin{bmatrix} d & 0 & 0 & f \\ 0 & c & e & 0 \\ 0 & e^* & b & 0 \\ f^* & 0 & 0 & a \end{bmatrix} \\
&= \begin{bmatrix} ad + |f|^2 & 0 & 0 & 2af \\ 0 & bc + |e|^2 & 2be & 0 \\ 0 & 2ce^* & bc + |e|^2 & 0 \\ 2df^* & 0 & 0 & ad + |f|^2 \end{bmatrix}.
\end{aligned}
\tag{A.6}
$$

To obtain the concurrence, we must use the eigenvalues of R: $\lambda_1 = ad + |f|^2(1 + 2\sqrt{ad})$, $\lambda_2 = ad + |f|^2(1 - 2\sqrt{ad})$, $\lambda_3 = bc + |e|^2(1 + 2\sqrt{bc})$, and $\lambda_4 = bc + |e|^2(1 - 2\sqrt{bc})$. Without specifying the matrix elements of ρ, we cannot go further. But the next steps are taking the square root of each eigenvalue of R, organizing them in decreasing order and substituting in Eq. (4.7).